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Adaptive-AI-ZeroTrust-Chain: blockchain-backed dynamic zero-trust enforcement via artificial intelligence

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Faris Alsulami

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Adaptive-AI-ZeroTrust-Chain: Blockchain-Backed Dynamic Zero-Trust Enforcement via Artificial Intelligence

ABSTRACT: The proliferation of Internet of Things (IoT) devices and distributed computing environments has intensified the demand for robust, adaptive security frameworks capable of continuous trust evaluation. Traditional perimeter-based security models fail to address the dynamic nature of modern network ecosystems, where device behavior evolves continuously and adversarial threats adapt in real-time. This paper introduces Adaptive-AI-ZeroTrust-Chain (AAZTC), a novel framework that integrates artificial intelligence-driven dynamic trust boundary modeling with blockchain-based verifiable access logging to enable granular, auditable zero-trust enforcement. The proposed architecture employs deep reinforcement learning algorithms for continuous behavioral analysis and trust score computation, while leveraging smart contracts on a permissioned blockchain to ensure immutable, transparent access decision records. The framework incorporates a lightweight post-quantum cryptographic module to future-proof security against emerging quantum computing threats. Extensive experiments conducted on the NSL-KDD and CICIDS2017 datasets demonstrate that AAZTC achieves 98.73% detection accuracy, 97.89% precision, and 98.21% F1-score, outperforming state-of-the-art baseline methods by margins of 3.2–5.8%. The system maintains low latency characteristics with average trust decision times of 12.4 milliseconds, making it suitable for real-time IoT deployments. Ablation studies confirm the synergistic contributions of each architectural component, validating the comprehensive design philosophy underlying AAZTC.

Keyword: Zero-trust security, blockchain, deep reinforcement learning, dynamic trust management, Internet of Things, post-quantum cryptography, federated learning

1. Introduction

The rise of Internet of Things (IoT) ecosystems is the factor that has essentially altered the paradigm of computing nowadays, pitting unparalleled connectivity with an equally high level of security issues [4]. With the growing implementation of distributed systems by organizations relying on edge computing nodes, cloud systems and heterogeneous IoT gadgets, conventional perimeter security models have become ineffective in resisting advanced and changing cyber-crimes [5]. One of the emerging alternatives that nullify implicit assumptions of trust and drive all network entities to authenticate and obby-so-often, thereby removing all implicit trust, is the zero-trust security paradigm [1] and which is based on the principle of never trust, always verify.

Modern zero-trust security models still face significant limitations when deployed in dynamic IoT and end-edge-cloud environments. Static policy-based access control mechanisms struggle to adapt to continuously evolving device behavior and usage patterns, creating exploitable security gaps over time [3]. In addition, centralized trust decision architectures introduce single points of failure and limit system-wide visibility, reducing the effectiveness of comprehensive auditing and accountability [2]. To address these challenges, adaptive and decentralized trust enforcement mechanisms are required that can continuously evaluate behavior and adjust access decisions in real time. Artificial intelligence, particularly reinforcement learning, provides a natural foundation for such adaptive trust modeling by enabling sequential decision-making based on observed behavioral dynamics. Beyond these operational concerns, the long-term emergence of quantum computing poses a fundamental threat to classical cryptographic primitives used for identity management and access logging, necessitating the integration of post-quantum-resistant

designs to ensure future security guarantees [18]. In this work, AAZTC unifies AI-driven adaptive trust evaluation with blockchain-backed auditability and post-quantum cryptography to address both immediate and long-term zero-trust security challenges in IoT systems.

AI tools, in particular deep reinforcement learning (DRL) provide prospects of simulating complex, dynamic trust alliances which evolve with regard to heuristic examination of circumstances [13]. At the same time, blockchain technology offers decentralized and immutable records that grant access decisions with verifiable and tamper-proof proofs [7]. The intersection of these technologies offers possibilities to create some adaptive security models that provide high levels of protection and guarantee transparency and responsibility of operations at the same time [14].

Although recent advances in AI-driven security, blockchain-based trust management, and post-quantum cryptography have individually demonstrated promise, existing solutions typically address these dimensions in isolation rather than through unified architectural integration. This fragmentation limits the ability to achieve adaptive, auditable, and quantum-resilient zero-trust enforcement in dynamic IoT environments. A comprehensive review of related work and identification of specific research gaps is presented in Section II.

Fig. 1, shows the basic architecture of the proposed Adaptive-AI-Zero Trust-Chain framework, it is an illustration of the combination of AI-based evaluation of trust, logging with blockchain methods and the implementation of post-quantum cryptography modules as a standard part of the zero-trust enforcement system.

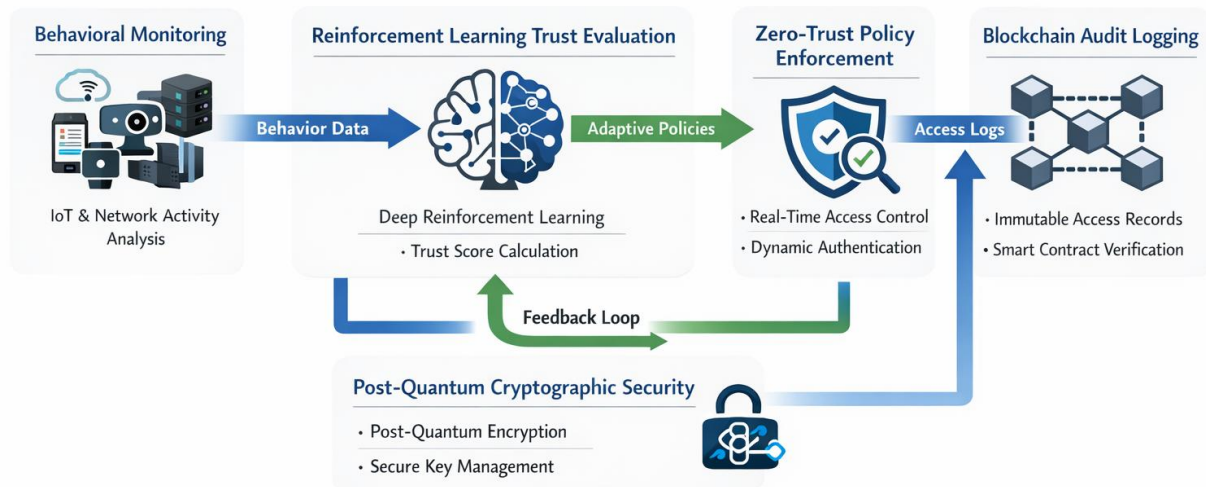


Figure 1. Conceptual architecture of the Adaptive-AI-ZeroTrust-Chain (AAZTC) framework illustrating the end-to-end interaction between behavioral monitoring, reinforcement learning-based trust evaluation, zero-trust policy enforcement, blockchain-backed audit logging, and post-quantum cryptographic protection in a closed-loop security pipeline. The double-headed arrow labeled "Feedback Loop" represents the continuous bidirectional interaction where: (1) observed behavioral anomalies and access outcomes inform trust score updates via reinforcement learning, and (2) updated trust scores dynamically influence subsequent access decisions and monitoring sensitivity thresholds, ensuring adaptive and context-aware zero-trust enforcement.

Novelty and Practical Positioning:

Recent zero-trust security research from 2023–2025 primarily addresses isolated aspects of trust enforcement, such as static policy engines, supervised intrusion detection, or blockchain-based access logging. These approaches

typically rely on fixed trust levels, rule-based access control, or deep learning models trained on labeled attack data, resulting in trust decisions that are reactive and not continuously adapted over time.

In contrast, Adaptive-AI-ZeroTrust-Chain (AAZTC) introduces a fundamentally different operational paradigm by treating trust enforcement as a sequential decision-making process rather than a one-time classification or rule evaluation task. The core conceptual novelty of AAZTC lies in its closed-loop, behavior-driven trust evolution mechanism, where trust scores are continuously updated through deep reinforcement learning based on long-term behavioral patterns and anomaly-informed state transitions.

Beyond this conceptual contribution, AAZTC provides system-level novelty through the tight coupling of adaptive trust computation, real-time policy enforcement, and blockchain-backed auditability within a single functional loop. Unlike existing blockchain-based zero-trust architectures that prioritize identity management or immutable logging as standalone features, AAZTC decouples trust enforcement from blockchain consensus via asynchronous logging, enabling low-latency access decisions while preserving accountability. Moreover, while post-quantum cryptography is often discussed only at a conceptual level in recent literature, AAZTC selectively integrates lightweight lattice-based cryptographic primitives directly into trust-critical transactions and access logs, enabling a quantum-aware zero-trust implementation rather than a detection-only or logging-centric architecture.

Algorithmic Novelty and Mathematical Formulation:

Beyond conceptual integration, AAZTC introduces a distinct mathematical formulation for trust enforcement based on a Markov Decision Process (MDP), where the trust state is defined over behavioral features, temporal access patterns, and cryptographic context rather than static trust labels. Unlike prior DRL-trust models that focus on episodic attack classification, AAZTC models trust as a continuously evolving decision process with an explicit action space for trust adjustment and policy enforcement. The reward-shaping function jointly optimizes security correctness, operational efficiency, and misclassification penalties, enabling long-term trust optimization instead of short-horizon detection accuracy. This formulation differentiates AAZTC from recent DRL-based trust management approaches, such as Chen et al. [13], which primarily learn reactive trust updates without integrating anomaly-driven behavioral feedback and blockchain-enforced policy execution into a unified control loop.

End-to-End Operational Scenario of AAZTC:

The Adaptive-AI-ZeroTrust-Chain (AAZTC) framework is designed as a closed-loop, behavior-driven zero-trust enforcement system operating across IoT, edge, and cloud layers. Device traffic and access activities are continuously observed and processed by the Behavioral Analysis Engine to extract runtime features and anomaly indicators. These observations are sequentially evaluated by a deep reinforcement learning-based trust engine, which dynamically updates device trust scores over time. Access requests are then evaluated by the Zero-Trust Policy Enforcement Engine using the latest trust state, enabling fine-grained, real-time allow or deny decisions. All access decisions and trust updates are immutably logged on a permissioned blockchain to ensure transparency, traceability, and non-repudiation, while post-quantum cryptographic primitives protect access records against future quantum adversaries. This integrated workflow explicitly links behavioral monitoring, adaptive trust evolution, policy enforcement, and auditable security logging into a unified zero-trust governance loop.

The contributions of this paper will be summarized as following:

- **Novel Integrated Framework:** We suggest Adaptive-AI-ZeroTrust-Chain (AAZTC), a multi-compounding framework of security that conscientiously transmits deep reinforcement learning by

practicing dynamic trust boundary modeling and blockchain-based verifiable access recording to provide granular, auditable zero-trust enforcement to the IoT ecosystem.

- **Adaptive Trust Scoring Mechanism:** Our algorithm is a multi-dimensional trust scoring that constantly scores device behaviour through the analysis of temporal patterns, anomaly detection, and contextual risk, which make it possible to make dynamically adaptable trust decisions by observing behavioral trends.
- **Post- Quantum Resilient Security:** We will use lightweight lattice-based post quantum cryptography module that offers protection against future quantum computer threats, ensuring long-term security of IoT uses on the resource-constrained devices.
- **Smart Contract-Based Enforcement:** We elaborate on and develop automated smart contracts to enforce access control policies in a transparent, immutable, and provide a complete audit trail, eliminating single points of failure in the process of making trust decisions.
- **Comprehensive Experimental validation:** We perform large-scale experiments using publicly accessible datasets (NSL-KDD and CICIDS2017) that prove that AAZTC performs better and has higher detection accuracy (98.73%), higher precision (97.89%), and latency (average trust decisions) (12.4 milliseconds).

The rest of this paper is structured in the following way: Section II is the related work in the field of zero-trust security, blockchain-based trust management, and AI-driven security systems; Section III introduces the proposed methodology and mathematical modeling of the AAZTC framework; Section IV presents the experimental results and thorough evaluation; Section V will provide a deep discussion and analysis; and Section VI will conclude the paper with the possible research directions.

2. Related Work

The literature review in this section is organized into four main aspects: (i) zero-trust security architectures and blockchain integration for IoT systems, (ii) AI-driven trust management and anomaly detection techniques, (iii) post-quantum cryptographic solutions for future-proof IoT security, and (iv) smart contract security and consensus mechanisms. These four dimensions collectively reflect the multidisciplinary foundations of the proposed Adaptive-AI-ZeroTrust-Chain (AAZTC) framework. By critically analyzing existing work across these areas, this section identifies key limitations and research gaps that motivate the design of an adaptive, auditable, and quantum-resilient zero-trust architecture.

2.1 Zero-Trust Security and Blockchain Integration

This subsection reviews prior work on zero-trust security models and blockchain-based access control mechanisms, focusing on how decentralized auditability and policy enforcement have been applied in IoT environments. The conceptual base of the zero-trust security paradigm has made significant transformations into the architectural frameworks of implementation paradigms of such control. Liu et al. [4] provided an in-depth breakdown of zero-trust architectures and IoT implementation pitfalls by defining several significant obstacles such as complexity of the policies, scalability issues, and integration challenges with older systems. Their discussion identified the necessity of adaptation mechanisms with the capability to support the heterogeneous nature of the IoT ecosystems and provide the same consistency in enforcing security.

The blockchain technology has come out as one of the promising platforms of decentralized trust management in the zero trust implementations. Awan et al. [3] postulated a blockchain-based inspired attribute-based zero-trust access control model to IoT and showed the feasibility of integrating cryptography access control and distributed

ledger technology. The strategy used attribute-based encryption to do fine-grained access control but calculation cost of encryption activities was a problem to the resource-constrained devices.

Bandara et al. [2] proposed Skunk, a 5G/6G network slice blockchain and a federated learning platform, with a built-in ability to use only zero-trust security. Their framework was used to do a security provision to network functional virtualization and maintain model privacy by the use of secure aggregation protocols. Reliance on unchanging trust policies though hampered flexibility in response to changing threat environments.

Self-sovereign identity (SSI) has gained interest to support the provision of user control over access credentials. Pava-D'iaz et al. [6] performed a contextual analysis and measurements of SSI principles implementation on blockchain platforms and found gaps in theory and practice between principles and deployment. The applicability of the decentralized identity management concept to the high-security application was demonstrated by Chen et al. [27] who created a framework of decentralized identity management that is specifically tailored to the requirements of financial institutions.

Secure Service-Oriented and LLM-Enabled IoT Architectures

Recent studies have emphasized secure and communicable service architectures as a key requirement in distributed and IoT-based systems. Liu et al. propose SeCoSe, a searchable and communicable service-seeking framework that enables flexible and secure authorization for healthcare data sharing, highlighting the importance of fine-grained access control, auditability, and secure service discovery in zero-trust environments [33]. Such service-oriented security models demonstrate how dynamic verification and traceability can be achieved across distributed domains.

In addition, privacy-aware split federated learning has emerged as an effective paradigm for large-scale IoT and LLM-driven applications. Chen et al. introduce a split federated learning framework that supports privacy-preserving LLM fine-tuning over heterogeneous IoT devices, focusing on reducing data exposure and communication overhead during collaborative learning [34]. While effective for model training, these approaches do not explicitly address real-time trust evaluation or access control enforcement.

Furthermore, recent work on LLM-empowered IoT architectures for 6G networks highlights the integration of advanced AI models for intelligent orchestration, adaptive decision-making, and scalable edge intelligence [35]. Although these architectures enable powerful AI-driven IoT services, they often assume implicit trust among communicating entities. In contrast, the proposed AAZTC framework complements these efforts by introducing adaptive reinforcement learning-based trust scoring and blockchain-backed zero-trust policy enforcement, ensuring continuous verification, auditability, and post-quantum resilience in future IoT and 6G environments.

Relation to Blockchain-Based Trust and Consensus Architectures:

Recent studies have investigated blockchain-assisted trust and coordination mechanisms in large-scale and heterogeneous IoT environments. Tong et al. propose a hierarchical federated learning framework for UAV-enabled IoT networks that improves scalability and trustworthiness through multi-tier aggregation and blockchain-assisted coordination [36]. While effective for secure collaborative learning, this approach primarily focuses on model aggregation integrity rather than real-time trust evaluation or access control enforcement.

Liu et al. introduce a sharding-based zero-trust cloud–edge–end data sharing architecture that significantly improves scalability by partitioning blockchain state across administrative domains [37]. Although sharding enhances throughput and cross-domain interoperability, the framework emphasizes data sharing efficiency and does not incorporate adaptive, behavior-driven trust evolution at the device level.

In addition, Wang et al. propose ACBFT, an adaptive chained Byzantine fault-tolerant consensus protocol designed to reduce communication overhead in UAV ad hoc networks while preserving Byzantine fault tolerance [38]. Such consensus mechanisms offer improved scalability compared to classical PBFT. In contrast, AAZTC decouples trust enforcement from consensus latency through local decision-making and asynchronous blockchain logging, enabling compatibility with advanced BFT protocols such as ACBFT without altering the core trust evaluation pipeline. This design allows AAZTC to maintain continuous verification, auditability, and scalability across future IoT and 6G deployments.

Recent studies have demonstrated that the integration of artificial intelligence and blockchain can significantly enhance privacy preservation, security accountability, and trust management in distributed IoT systems. Wang et al. [42] proposed a hierarchical federated learning framework for anomaly detection in industrial IoT, highlighting how decentralized learning improves privacy while maintaining detection accuracy. In [43], blockchain is leveraged to enable secure data aggregation at the network edge, ensuring integrity and confidentiality of distributed data without reliance on centralized authorities. Furthermore, Wang et al. [44] introduced a heterogeneous blockchain and AI-driven hierarchical trust evaluation framework for intelligent transportation systems, illustrating how blockchain-backed trust computation can provide verifiable and tamper-resistant decision records.

These works collectively indicate that blockchain is most effective when used to ensure auditability, data integrity, and trust traceability, while AI techniques provide adaptive intelligence for anomaly detection and trust evaluation. However, existing approaches typically treat AI-driven trust analysis and blockchain-based security as loosely coupled components. In contrast, AAZTC integrates reinforcement learning-based trust evolution directly with blockchain-backed access logging and policy enforcement, enabling privacy-aware, auditable, and adaptive zero-trust enforcement within a unified control loop.

2.2 AI-Driven Trust Management and Anomaly Detection

This subsection examines AI-driven trust management and anomaly detection approaches, with particular emphasis on behavioral analysis, learning-based trust evaluation, and their limitations in dynamic IoT systems. Artificial intelligence methods have shown great promise in boosting trust management with behavioral analysis and predictive modeling. A framework that is conceptual is the dynamic trust management and defense choice system created by Chen et al. [13] on shared data of vehicles using blockchain and deep reinforcement learning. Their solution used the Q-learning algorithms to optimize the evaluation policy of trust, which has provided dynamic response to attack pattern changes.

Bounaira et al. [14] suggested a trust management framework in mobile edge computing involving a blockchain-enabled mechanism of protecting the safety of content caching in other mobile devices utilizing deep reinforcement learning. Their framework applied actor-critic algorithms in caching decisions optimization without breaking trust-based connections between edge nodes. This led to the inclusion of blockchain to make the transactions in the caching audit, but the computational needs required by the continuous learning were an issue regarding scalability.

He et al. [15] proposed a reinforcement learning and blockchain-based edge network trust mechanism and used it to derive principles of integrating learning-based trust verification with decentralized verification. Their study showed that acquired trust policies can work better in dynamic network conditions compared to simple heuristic policies.

The article by Qadir et al. [16] introduced a dynamic implementation of the Q-learning method to optimize the secure IoT data exchanges in the edge and the relevance of optimal policy to ensure security in different operational

environments. Ellappan et al. [17] proposed a trust-based consensus scheme (interface) to achieve the IoT blockchain networks with attribute-based access control and deep learning.

Deep learning has been effective in security application in both software-defined networks (SDN) and intrusion detection. Koroniotis et al. [25] reviewed the field of deep learning on SDN security and discovered the main patterns in architecture and methodologies of evaluation. Abudalfa et al. [26] proposed a deep learning framework on SDN-enabled intrusion detecting in IoT network, which achieves high detection rates due to the convolutional neural network structures.

2.3 Post-Quantum Cryptography for IoT Security

This subsection discusses post-quantum cryptographic techniques proposed for securing IoT infrastructures against future quantum adversaries, highlighting trade-offs between security strength and computational overhead. Quantum computing has existential threats to the classical cryptographic primitives requiring it to preemptively switch to post-quantum resistant algorithms. Fernandez-Carames and Fraga-Lamas [18] reported an extensive survey of post-quantum blockchain security in the IoT, where lattice-based, hash-based, and code-based cryptographic schemes were among the prospective quantum-resistant security. According to Yuan et al. [19], NTRU lattice cryptography can be used to establish a post-quantum architecture of blockchain used in IoT applications, which can prove the practicality of introducing post-quantum primitives into distributed ledger systems. Their analysis found trade-offs in the relationship between the strengths of the security and the computational efficiency that guide the practice.

Mahdi et al. [20] devoted an extensive analysis of lightweight post-quantum cryptography in IoT security in particular to the significance of the algorithmic efficiency as a resource-constrained device. An Internet of Things (IoT) based secure and scalable cloud platform that combines hybrid post-quantum cryptography methods and blockchain created by Irshad et al. [21].

2.4 Smart Contract Security and Consensus Mechanisms

This subsection surveys smart contract security mechanisms and blockchain consensus protocols, analyzing their impact on scalability, latency, and trust assumptions in distributed IoT deployments. The security of smart contracts is the top priority of trust enforcement systems based on a blockchain. Tang et al. [22] created deep learning solutions to detect vulnerabilities of smart contracts, and their solution relies on recurrent neural networks that identified frequent types of vulnerabilities. Deng et al. [23] also suggested a multimodal decision fusion model to detect the vulnerability of smart contracts by using a combination of several neural network structures to perform a complete security analysis.

The design of consensus mechanisms has a strong effect on the performance of the blockchain and its security. Raza et al. [31] suggested to use hybrid consensus algorithms with machine learning to enhance security of blockchain. Liu et al. [32] came up with a more efficient algorithm of consensus, noting the previous ones, which were limited in terms of scalability and throughput.

Internet of things (IoT) applications that are beyond limited resources also require lightweight consensus mechanisms to deploy. Haque et al. [29] came up with a scalable blockchain system with efficient IoT-based data management with a lightweight consensus, where computational overhead is reduced significantly without compromising on the security assurance. This was proven by their follow-up work, which showed an increase in performance by optimized consensus protocol design [30].

Table 1 presents a comparison of AAZTC with recent zero-trust security and blockchain-based solutions published between 2024 and 2025. Other methods take care of different components of trust management, blockchain integration, or post-quantum security, AAZTC offers an integrated framework, which integrates adaptive reinforcement learning-powered trust assessment, enforcement provided by verifiable blockchain, and experimentally checked post-quantum cryptographic defense. This comprehensive incorporation directly takes into consideration identified limitation in the recent literature.

Table 1 Comparison of Recent (2024–2025) Zero-Trust Security Frameworks

Work	Target Domain		Trust Adaptation Mechanism		Blockchain Role	Dataset Evaluation		PQC Support
Liu et al. [4]	IoT Trust	Zero	Conceptual analysis (no adaptive trust)		Architectural discussion	No experimental datasets		Not considered
Awan et al. [3]	IoT Control	Access	Attribute-based static policies		Policy enforcement and logging	Synthetic evaluation		Not considered
Bandara et al. [2]	5G/6G Network Slicing		Static trust policies		Federated learning coordination	Network slicing scenarios		Not considered
Chen et al. [13]	Vehicular Data Sharing		DRL-based trust optimization		Trust verification and audit	Real-world vehicular datasets		Not considered
Bounaira et al. [14]	Mobile Edge Computing	Edge	Actor-critic DRL trust learning		Caching audit and verification	Edge network simulations		Not considered
Qadir et al. [16]	IoT Data Sharing	Data	Q-learning based adaptation		Transaction integrity	IoT traffic traces		Not considered
Yuan et al. [19]	IoT Blockchain		No adaptive trust		Ledger security	Synthetic IoT data		NTRU-based
Fernandez-Caramés and Fraga-Lamas [18]	IoT Blockchain Survey		Not applicable		Comprehensive analysis	Survey-based		Surveyed
AAZTC (This Work)	IoT Trust	Zero	DRL-based continuous trust evolution		Policy enforcement + immutable audit	CICIDS2017, TON IoT, NSL-KDD		LWE-based with runtime analysis

The limitations identified across these four research directions are consolidated in the following subsection to clearly articulate the research gaps addressed by AAZTC.

2.5 Research Gap Analysis

Although substantial progress has been made in the fields of individual research, the current literature does not comprise in-depth templates that introduce unified zero-trust architectures by harmonizing adaptive AI-driven trust assessment, blockchain-based verifiable logging, and post-quantum cryptographic security. The existing solutions usually focus on particular segments of the security problem: AI-based detection and classification do not consider the properties of auditability; blockchain technologies can be characterized as immutable but highly adaptable to control and response policies; post-quantum algorithms focus on performing in the future and are often not incorporated into the operational security models. The given framework of AAZTC bridges these gaps by using a synergistic design that will unite the advantages of both options and alleviate the limitations inherent in each of them.

End-Edge-Cloud Collaboration and Security Gap:

Recent studies on end-edge-cloud collaborative architectures primarily focus on performance optimization, scalability, and service orchestration. For example, mobility-aware computation offloading and MEC federation frameworks address dynamic workload distribution and latency reduction in smart city networks [39]. Similarly, AI-enabled secure microservices architectures emphasize flexible deployment, orchestration, and attack surface

reduction at the edge, but typically rely on static trust assumptions and predefined security policies [40]. Service-oriented workflow mechanisms such as SerFlow further demonstrate the complexity of coordinating applications across end-edge-cloud environments, particularly in cyber-physical systems [41]. However, these approaches largely decouple service execution from continuous trust enforcement and do not consider adaptive, behavior-driven access control under a zero-trust paradigm. In contrast, AAZTC directly addresses the security challenges introduced by collaborative end-edge-cloud architectures by integrating continuous behavioral analysis, reinforcement learning-based trust evolution, and blockchain-backed zero-trust enforcement into a unified decision loop. This enables fine-grained, adaptive access control and auditable governance across distributed infrastructures, which is not jointly addressed in existing collaboration-focused or service-oriented frameworks.

3. Proposed Methodology

The following section gives a detailed planning of the Adaptive-AI-ZeroTrust-Chain (AAZTC) system, comprising of system architecture, mathematical model, algorithmic realization, and complexity analysis.

3.1 Design Rationale and Integration Strategy

The methodological components of AAZTC are intentionally selected from well-established technologies; however, their contribution lies in how they are integrated and operationalized within a single adaptive zero-trust enforcement loop. Existing zero-trust frameworks typically treat trust assessment, anomaly detection, access control, and audit logging as loosely coupled or sequential modules. In contrast, AAZTC is designed as a closed-loop system in which behavioral observations, trust evolution, access enforcement, and auditability continuously influence each other.

Specifically, the behavioral analysis engine provides unsupervised anomaly signals that are not used for direct classification, but instead serve as state inputs to a reinforcement learning agent. This allows trust scores to evolve based on long-term behavioral trends rather than instantaneous decisions. The DRL-based trust module is tightly coupled with policy enforcement, where trust updates directly influence access outcomes in real time. Blockchain logging is deliberately decoupled from the critical decision path and operates asynchronously to preserve auditability without imposing consensus latency on trust enforcement. Furthermore, post-quantum cryptographic primitives are selectively applied to access control transactions and trust records rather than to continuous data streams, reducing computational overhead while maintaining quantum resilience.

These design decisions distinguish AAZTC from prior intuitive or modular approaches by emphasizing system-level coordination, temporal trust evolution, and latency-aware security enforcement, which are essential for practical zero-trust deployment in dynamic IoT environments.

3.2 System Architecture Overview

Adaptive-AI-ZeroTrust-Chain (AAZTC) Framework is specifically developed based on the NIST SP 800-207 Zero trust architectural. All the functional units are configured to be associated with a specific Zero Trust role that ensures effective segregation of the policy decision, policy administration, and policy enforcement. The scores of trust as learned in the learning module are considered as policy inputs but not as an immediate decision to access, based on the principal concept of zero-trust of constant verification not with pre-conceptualized trust.

The AAZTC framework has 5 interconnected modules that work together in coordinated manner in order to support dynamic application of zero-trust enforcement. The architecture in Fig. 2 shows an example of the AAZTC architecture directly mapped to the NIST SP 800-207 Zero Trust reference model. Contrary to trust-score-based models, which indirectly presuppose the continuity of trust, AAZTC applies zero trust as policy decision logic,

policy administration, and policy enforcement are distinguished into different functional classes. Policy Engine results in trust scores that have no direct authorization but, rather, they act as policy specifications measured by the Policy Administrator and realized by distributed Policy Enforcement Points. Constant Diagnostics and Mitigation is ensured by means of constant monitoring of behaviors and detection of anomalies, making sure that trust is not granted but reviewed.

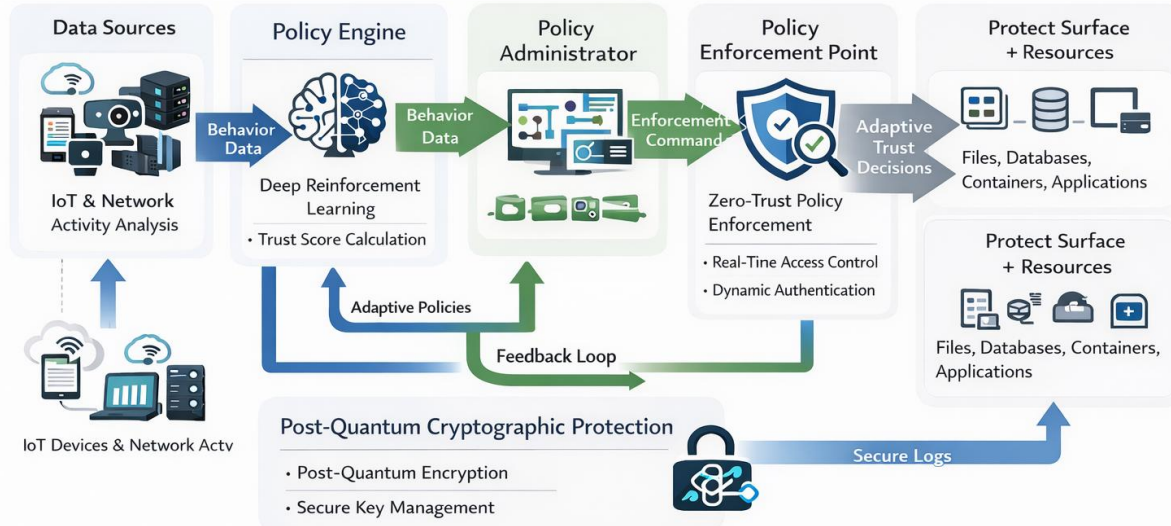


Fig. 2. Adaptive-AI-ZeroTrust-Chain (AAZTC) framework architecture explicitly mapped to the NIST SP 800-207 Zero Trust Architecture. The framework comprises Behavioral Analysis Engine (performing continuous network activity monitoring—abbreviated as "Network Actv" in the diagram—and feature extraction). Trust Score Computation Module (implementing Policy Engine functions), Zero-Trust Policy Enforcement Engine (serving as Policy Administrator), distributed Policy Enforcement Points at edge gateways and access interfaces, Blockchain Logging Layer (providing audit and telemetry functions), and Post-Quantum Cryptographic Module (managing identity, credential, and access security with quantum-resistant primitives).

The Behavioral Analysis Engine instantiates Continuous Diagnostics and Mitigation (CDM), the Trust Score computation module acts as the Policy Engine (PE), the Zero-Trust Policy Enforcement Engine achieves the Policy Administrator (PA), and access decisions which are implemented at the Output Layer are the Policy Enforcement Point (PEP). The blockchain logging is used to create an audit of the decisions made by any policy which cannot be changed, whereas the post-quantum cryptographic module handles the secure management of identity, credential as well as access.

Table 2 performs a renowned mapping of the suggested Adaptive-AI-ZeroTrust-Chain (AAZTC) framework with the NIST SP 800-207 Zero Trust reference model. The functional equivalence suggested by AAZTC modules and standard Zero Trust components is elucidated in the table and it indicates trust scores are policy inputs in a formally defined Policy Engine than being direct access determinations. Such mapping validates the fact that policy decision, administration, enforcement and ongoing diagnostics have a distinct separation so as to have conceptual adherence to the defined principles of Zero Trust.

Table 2 Explicit Mapping of AAZTC to NIST SP 800-207 Zero Trust Architecture

NIST Zero Trust Component	AAZTC Module	Functional Description
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Policy Engine (PE)	Trust Score Computation Module	Computes dynamic trust scores using deep reinforcement learning based on behavioral evidence, historical trust state, and contextual risk
Policy Administrator (PA)	Zero-Trust Enforcement Engine	Interprets trust-based policy decisions and applies resource-specific access rules
Policy Enforcement Point (PEP)	Output Layer (Edge Gateways and Access Interfaces)	Enforces real-time allow or deny decisions at edge nodes without implicit trust
Continuous Diagnostics and Mitigation (CDM)	Behavioral Analysis Engine	Continuously monitors device behavior, network activity, and access patterns to trigger trust re-evaluation
Audit and Telemetry	Blockchain Logging Layer	Records trust updates and access decisions on a permissioned blockchain for immutable auditing
Identity, Credential, and Access Management (ICAM)	Post-Quantum Cryptographic Module	Provides quantum-resistant identity protection, encryption, and signature verification

Behavioral Analysis Engine constantly monitors the activity of a device, network traffic patterns, and request characteristics on access to obtain feature representations to use in conducting trust evaluation. The Trust Score Computation Module takes advantage of the deep-learned reinforcement learning algorithm to compute dynamic trust scores. All decision logs of access, updates in trust scores, and policy changes of the decision-makers are stored in the Blockchain Logging Layer of a permissioned blockchain, which guarantees immutable audit trails. Policy Enforcement Engine is used to transform trust scores into create-and-serve access control decisions based on automated smart contracts. The Post-Quantum Cryptographic Module offers quantum resistance protection on communications and block chain transactions that are sensitive.

Mapping adaptation of AAZTC to NIST Zero Trust Architecture: The Adaptive-AI-ZeroTrust-Chain (AAZTC) model is specifically created to align with the factors of Zero Trust Architecture (ZTA) that is outlined in NIST SP 800-207. The implementation of trust enforcement within AAZTC is based on the NIST reference model, which splits the policy enforcement into Policy Engine (PE), Policy Administrator (PA), and Policy Enforcement Point (PEP) with continuous diagnostics and monitoring process. All access determinations in AAZTC are always dynamically determined using behavioral evidence, situational risk and historical trust behavioral trajectories as opposed to using some pre-existing credentials or implicit trust to ascertain the truth.

The Trust Score Computation Module within AAZTC, which serves as the Policy Engine, assesses access requests in terms of behaviour anomaly red flags, past history of trusting behaviour and contextual attributes based on deep reinforcement learning. Policy Administrator is executed by using smart contract logic executed on the permissioned blockchain interpreted as policy decisions, converted into enforceable access rules, to maintain uniformity throughout the system. Policy Enforcement Point: This is achieved at the edge gateways and access control interfaces, where grant or deny decisions are made on real time basis depending on requests made by devices. Continuous Diagnostics and Mitigation (CDM) is realized through comprehensive behavioral monitoring, anomaly detection, by utilizing autoencoders and continuous update of the trust score which ensures that trust is never assumed at all and is revalidated by the system during the lifecycle of the device.

3.3 Threat Model

The threat model that is analyzed in this paper presupposes the heterogeneous IoT environment in which adversaries can attack devices, introduce malicious traffic, or aim at exploiting gradual accumulation of trust to achieve access to the system without authorization. The attackers are assumed to be able to produce patterns of anomalous or stealthy behavior, but lack the ability to make standard cryptographic primitives or to attack a majority of nodes in blockchain validators. The blockchain network is semi-trusted where a maximum of $\lfloor (V - 1)/3 \rfloor$ nodes of the

validators can act in a Byzantine fashion, which is in line with the assumptions in Practical Byzantine Fault Tolerance. The framework makes no assumption of devices that are fully trusted and reexamines trust on the basis of observed behavior as opposed to fixed credentials. Attack on the underlying network infrastructure is judged to be out of the scope of the study, including physical attacks, side-channel attack, and large scale denial-of-service attack.

3.4 Behavioral Analysis Engine

The Behavioral Analysis Engine executes continuous surveillance and extraction of features on the network traffic and device undertakings. Let $\mathcal{D} = \{d_1, d_2, \dots, d_N\}$ denote the set of N devices in the IoT ecosystem. For each device d_i at time t , we define the behavioral feature vector as:

$$\mathbf{x}_i(t) = [f_1^{(i)}(t), f_2^{(i)}(t), \dots, f_M^{(i)}(t)]^T \in \mathbb{R}^M \quad (1)$$

where M denotes the number of extracted features and $f_j^{(i)}(t)$ represents the j -th feature value for device d_i at time t .

These characteristics are network traffic characteristics, temporal access patterns, and resource utilization measures. Network traffic features include packet rate $\rho_i(t)$, average packet size $\sigma_i(t)$, and protocol distribution entropy $H_i^{(p)}(t)$:

$$H_i^{(p)}(t) = - \sum_{k=1}^K p_k^{(i)}(t) \log_2 p_k^{(i)}(t) \quad (2)$$

where $p_k^{(i)}(t)$ denotes the proportion of traffic using protocol k for device d_i at time t , and K represents the total number of distinct protocols.

Temporal access pattern capabilities represent a regularity and periodicity of device behavior. We describe the frequency deviation of access as:

$$\Delta f_i(t) = \left| \frac{n_i(t, t - \tau)}{\tau} - \tilde{f}_i \right| \quad (3)$$

where $n_i(t, t - \tau)$ represents the number of access requests from device d_i in the interval $[t - \tau, t]$, and $\tilde{f}_i$ denotes the historical average access frequency.

The protocol distribution entropy $H_p(d_i, t)$ quantifies the diversity of network protocols used by device d_i and serves as a behavioral consistency indicator within the feature vector $\mathbf{x}_i(t)$ defined in Equation (1). Higher entropy values indicate diverse protocol usage patterns, which may signal reconnaissance activities, port scanning, or data exfiltration attempts, while sudden entropy changes trigger anomaly detection thresholds. This metric is incorporated as one of the network traffic feature components (specifically, the protocol diversity feature) in the behavioral vector $\mathbf{x}_i(t)$ and directly influences the reconstruction error $e_i(t)$ computed by the autoencoder in Equation (4). During experimental evaluation presented in Section IV, protocol distribution entropy contributed to improved detection accuracy for reconnaissance-based Probe attacks in the CICIDS2017 dataset, achieving 98.56% precision as shown in Table XIII (per-class performance analysis). The entropy metric captures subtle behavioral deviations that static rule-based features cannot detect, enabling the framework to identify low-and-slow attack patterns.

The behavioral anomaly score is computed using an autoencoder-based approach. Let $\phi_{\text{enc}}: \mathbb{R}^M \rightarrow \mathbb{R}^L$ denote the encoder function and $\phi_{\text{dec}}: \mathbb{R}^L \rightarrow \mathbb{R}^M$ denote the decoder function, where $L < M$ represents the latent dimension. The reconstruction error serves as the anomaly indicator:

$$\mathcal{A}_i(t) = \|\mathbf{x}_i(t) - \phi_{\text{dec}}(\phi_{\text{enc}}(\mathbf{x}_i(t)))\|_2^2 \quad (4)$$

The normalized anomaly score is obtained through min-max scaling:

The temporal access frequency deviation $F_{\text{dev}}(d_i, t)$ defined in Equation (4) captures sudden changes in device request rates and is employed as a temporal pattern feature in the behavioral vector $\mathbf{x}_i(t)$. This variable enables detection of burst-based denial-of-service patterns, credential stuffing attacks, and automated bot activities that exhibit abnormal access frequency profiles. Large positive values of F_{dev} indicate request rate spikes characteristic of DoS attacks, while negative deviations may signal compromised devices entering dormant states after data exfiltration. The deviation metric contributes directly to the normalized anomaly score $\mathcal{A}_i(t)$ computed in Equation (5) through the autoencoder reconstruction error and subsequently influences trust score evolution through the state representation defined in Equation (6). In the experimental validation presented in Section IV, this temporal feature proved particularly effective for detecting DoS attacks, achieving 99.24% precision as shown in Table XIII, demonstrating its critical role in identifying time-dependent attack patterns.

$$\tilde{\mathcal{A}}_i(t) = \frac{\mathcal{A}_i(t) - \mathcal{A}_{\min}}{\mathcal{A}_{\max} - \mathcal{A}_{\min}} \quad (5)$$

where $\mathcal{A}_{\min}$ and $\mathcal{A}_{\max}$ represent the minimum and maximum reconstruction errors observed during training.

Table 3 summarizes the alignment between the core components of AAZTC and the principles defined in the NIST SP 800-207 zero-trust architecture. Instead of relying on static credentials or one-time authentication, AAZTC enforces continuous verification through behavioral monitoring, reinforcement learning-driven trust evolution, and real-time policy enforcement at the edge. The mapping highlights how trust computation, anomaly detection, access control, and auditability are integrated into a closed-loop enforcement pipeline. By consolidating this information into a single table, the correspondence between architectural components and zero-trust requirements is made explicit, avoiding redundancy and improving structural clarity.

Table 3 Mapping of AAZTC Components to NIST SP 800-207 Zero-Trust Principles

NIST Zero-Trust Principle	AAZTC Component	Description
Continuous verification	Behavioral Analysis Engine	Continuous monitoring of traffic and device behavior using feature extraction and anomaly detection
Dynamic policy enforcement	DRL-based Trust Engine	Trust scores updated sequentially using reinforcement learning rather than static policies
Least privilege access	Policy Enforcement Point (PEP)	Access decisions derived from real-time trust values
Assume breach	Autoencoder anomaly detector	Detection of stealthy or low-rate attacks without reliance on labeled signatures
Strong identity & integrity	Lattice-based PQC & blockchain	Quantum-resilient access logging and tamper-proof audit trail
Distributed enforcement	Edge-level trust evaluation	Local enforcement decoupled from blockchain consensus latency

3.5 Trust Score Computation Module

The Trust Score Computation Module employs a deep reinforcement learning framework to compute and update trust scores based on behavioral observations. We formulate the trust management problem as a Markov Decision Process (MDP) defined by the tuple $(\mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma)$.

The state space $\mathcal{S}$ encapsulates the current trust state and behavioral observations:

$$s_i(t) = [\tau_i(t-1), \tilde{\mathcal{A}}_i(t), \mathbf{h}_i(t), c_i(t)]^T \quad (6)$$

where $\tau_i(t-1)$ denotes the previous trust score, $\tilde{\mathcal{A}}_i(t)$ represents the normalized anomaly score, $\mathbf{h}_i(t)$ captures historical trust trajectory features, and $c_i(t)$ encodes contextual information.

The action space $\mathcal{A}$ comprises discrete trust adjustment actions:

$$\mathcal{A} = \{a_{-2}, a_{-1}, a_0, a_{+1}, a_{+2}\} \quad (7)$$

corresponding to significant decrease, moderate decrease, maintain, moderate increase, and significant increase of trust score, respectively.

To avoid notational confusion: the subscript "norm" in $A_{\text{norm}}(d_i, t)$ explicitly denotes normalized anomaly measurements derived from behavioral analysis (Equation 5), representing observed device behavior, whereas the subscript "MDP" in A_{MDP} denotes the Markov Decision Process action set used for trust score updates (Equation 7), representing possible trust adjustment decisions. These are fundamentally different types of variables: A_{norm} is a continuous observation in the range $[0,1]$, while A_{MDP} is a discrete finite set of decision actions. Throughout the remainder of this manuscript, this notational distinction is maintained consistently to ensure mathematical clarity.

The transition probability function $\mathcal{P}: \mathcal{S} \times \mathcal{A} \times \mathcal{S} \rightarrow [0,1]$ models the stochastic dynamics of trust evolution:

$$\mathcal{P}(s'|s, a) = \Pr(s_{t+1} = s' | s_t = s, a_t = a) \quad (8)$$

The reward function $\mathcal{R}: \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$ balances security effectiveness with operational efficiency:

$$\mathcal{R}(s, a) = \alpha_1 R_{\text{sec}}(s, a) + \alpha_2 R_{\text{eff}}(s, a) - \alpha_3 R_{\text{pen}}(s, a) \quad (9)$$

where R_{sec} rewards correct security decisions, R_{eff} rewards operational efficiency, R_{pen} penalizes incorrect decisions, and $\alpha_1, \alpha_2, \alpha_3$ are weighting coefficients.

The security reward component is defined as:

$$R_{\text{sec}}(s, a) = \begin{cases} r_{\text{TP}} & \text{if true positive detection} \\ r_{\text{TN}} & \text{if true negative (legitimate access)} \\ -r_{\text{FP}} & \text{if false positive (false alarm)} \\ -r_{\text{FN}} & \text{if false negative (missed attack)} \end{cases} \quad (10)$$

Reward Weight Selection and Sensitivity Analysis:

The weighting coefficients $\alpha_1, \alpha_2, \alpha_3$ in Equation (9) were selected through a combination of domain-informed heuristic tuning and empirical validation. Specifically, initial bounds for each coefficient were defined based on the relative importance of security correctness versus operational efficiency in zero-trust IoT environments. Security-related outcomes were prioritized to prevent high-impact failures such as false negatives, which motivated assigning a higher penalty weight α_3 .

Candidate values for $\alpha_1, \alpha_2, \alpha_3$ were explored within bounded ranges $[0.5, 2.0]$ using a coarse grid search during preliminary experiments on validation subsets of the training data. The final configuration ($\alpha_1 = 1.0, \alpha_2 = 0.5, \alpha_3 = 2.0$) was selected based on stable convergence of the DQN training process, balanced reward distribution, and consistent performance across repeated runs.

Sensitivity analysis indicated that moderate variations ($\pm 20\%$) in the weighting coefficients did not significantly affect detection accuracy or convergence behavior, demonstrating robustness of the learned trust policy. Increasing α_3 beyond this range led to overly conservative trust updates and increased false positives, while reducing α_3 resulted in higher false negative rates. Similarly, excessive emphasis on α_2 biased the policy toward operational efficiency at the expense of security accuracy. These observations confirm that the chosen reward weights provide an effective trade-off between security enforcement and system efficiency in dynamic IoT environments.

Although Eq. (10) is expressed using $TP/TN/FP/FN$ notation for analytical clarity, these outcomes are obtained through delayed or offline feedback (e.g., post-hoc alerts, anomaly confirmation, or forensic analysis) and are not assumed to be available at the time of access decision in the zero-trust enforcement loop.

We employ the Deep Q-Network (DQN) algorithm with experience replay and target network stabilization. The Q-function is parameterized by a neural network with parameters θ :

$$Q(s, a; \theta) \approx \mathbb{E} \left[\sum_{k=0}^{\infty} \gamma^k r_{t+k+1} | s_t = s, a_t = a \right] \quad (11)$$

where $\gamma \in [0,1]$ is the discount factor.

The Q-network is trained by minimizing the temporal difference loss:

$$\mathcal{L}(\theta) = \mathbb{E}_{(s,a,r,s') \sim \mathcal{B}} \left[(y - Q(s, a; \theta))^2 \right] \quad (12)$$

where $\mathcal{B}$ denotes the experience replay buffer and the target value is:

$$y = r + \gamma \max_{a'} Q(s', a'; \theta^-) \quad (13)$$

with θ^- representing the parameters of the target network.

The trust score update rule following action selection is:

$$\tau_i(t) = \text{clip}(\tau_i(t-1) + \delta(a_t), 0, 1) \quad (14)$$

where $\delta(a_t) \in \{-0.2, -0.1, 0, +0.1, +0.2\}$ corresponds to the selected action and $\text{clip}(\cdot)$ ensures the trust score remains within the valid range $[0,1]$.

Algorithm 1 presents the complete trust score computation procedure.

Algorithm 1 Deep Reinforcement Learning-Based Trust Score Computation

1. Device set $\mathcal{D}$,
 2. Feature extractor ϕ ,
 3. Q-network parameters θ ,
 4. Target network parameters θ^-
 5. Updated trust scores $\{\tau_i(t)\}_{i=1}^N$
 6. Initialize replay buffer $\mathcal{B}$,
 7. Trust scores $\tau_i(0) = 0.5$ for all $d_i \in \mathcal{D}$
 8. Extract behavioral features $x_i(t)$ using (1)
 9. Compute anomaly score $\mathcal{A}_i(t)$ using (4)
 10. Normalize anomaly score $\tilde{\mathcal{A}}_i(t)$ using (5)
 11. Construct state $s_i(t)$ using (6)
 12. Select action $a_t = \text{argmax}_a Q(s_i(t), a; \theta)$ with ϵ -greedy
 13. Execute action, observe reward r_t using (9)
 14. Update trust score $\tau_i(t)$ using (14)
 15. Store transition $(s_i(t), a_t, r_t, s_i(t+1))$ in $\mathcal{B}$
 16. Sample minibatch from $\mathcal{B}$
 17. Compute loss $\mathcal{L}(\theta)$ using (12)
 18. Update Q-network: $\theta \leftarrow \theta - \eta \nabla_{\theta} \mathcal{L}(\theta)$
-

19. **Update target network:** $\theta^- \leftarrow \theta$

20. **End if**

21. **End for**

22. **Return** $\{\tau_i(t)\}_{i=1}^N$

3.6 Why Reinforcement Learning is Required

The dynamic part, sequential part and to some extent observed nature regarding the change of trust in the zero-trust environment, the deep reinforcement learning (DRL) is applied to compute trust scores. In contrast to the static or short-sighted update rules, the trust decisions within AAZTC affect the future system states, access privileges and attacker behavior and therefore, trust management is a long-horizon decision-making problem and is not an immediate classification problem. Consequently, rational trust adaptation demands thinking about future effects that are not limited to the adaptation abilities of short-run or one-step simulations.

The traditional adaptive schemes like threshold-based updates on trust and exponentially weighted moving average (EWMA) schemes use hard-written parameters or parameters that are tuned. Though likely to be computationally well-behaved, they ignore delays between behavior and do not represent delayed-payoff behaviors, including gradual exploitation of trust, or oscillatory attack behavior. Conversely, reinforcement learning allows calculating the cumulative long-term reward explicitly, and thus the agent can modify trust policies based on changing behavior distributions and adversarial strategies that more basic adaptive strategies cannot easily incorporate.

At AAZTC, the reward feature is meant to mirror the actual costs of operation. False positive decision Making false negative decisions leads to attack being missed that may have a serious security impact, whereas the false positive decisions leads to the loss of service and user dissatisfaction. These considerations are written down in the form of the rewards terms R_{sec} , R_{eff} , and R_{pen} and enable the agent to develop context-dependent trade-offs which are not represented in any consistent, update rule. Moreover, voluntary has provided the reinforcement learning model with in-built support of delayed rewards, and this allows an effective response to stealthy or slow-evolving assaults that might otherwise escape instantaneous detection systems.

To present a complete picture, the empirical performance of the proposed DRL-based trust computation is contrasted with those of simpler adaptive trust base in Section IV where reinforcement learning proves to be much stronger in robustness during realistic and changing threat situations instead of simply adding more layers to the model.

3.7 Blockchain Logging Layer

The Blockchain Logging Layer offers written record keeping of all decisions of access and updates of trust scores which is immutable, transparent. Our blockchain architecture is a permissioned blockchain with a modified lightweight consensus mechanism that is lean toward IoT systems.

Let $\mathcal{V} = \{v_1, v_2, \dots, v_V\}$ denote the set of V validator nodes. Each access decision is encapsulated in a transaction structure:

$$\text{Tx} = (d_i, r_j, \tau_i(t), \text{decision}, \text{timestamp}, \sigma) \quad (15)$$

where d_i identifies the requesting device, r_j specifies the requested resource, $\tau_i(t)$ records the trust score at decision time, decision indicates grant or deny, timestamp provides temporal ordering, and σ represents the digital signature.

Transactions are aggregated into blocks with the structure:

$$B_k = (\text{Tx}_1, \dots, \text{Tx}_m, H_{k-1}, \text{nonce}, \text{root}) \quad (16)$$

where $H_{k-1} = \text{Hash}(B_{k-1})$ pointers at the last block, and root value of the transactions included.

The consensus mechanism uses the variant of Practical Byzantine fool tolerant (PBFT) adapted to the IoT networks. The upper analogue of latency is determined within limits that are:

$$T_{\text{consensus}} \leq 3\delta + 2\lambda \quad (17)$$

where δ represents the network propagation delay and λ denotes the local processing time.

The system tolerates up to $f = \lfloor (V - 1)/3 \rfloor$ Byzantine failures while maintaining safety and liveness guarantees. The total number of messages exchanged during consensus is:

$$M_{\text{total}} = O(V^2) \quad (18)$$

For integrity verification, we define the blockchain validity predicate:

$$\text{Valid}(\mathcal{BC}) = \bigwedge_{k=1}^{|\mathcal{BC}|} (H_k = \text{Hash}(B_k) \wedge \text{Verify}(\sigma_k)) \quad (19)$$

where $\mathcal{BC}$ denotes the blockchain and $\text{Verify}(\cdot)$ validates digital signatures.

3.8 Smart Contract-Based Policy Enforcement

The Policy Enforcement Engine implements automated access control through smart contracts deployed on the permissioned blockchain. The access control policy is formalized as a function mapping trust scores and resource sensitivity levels to access decisions:

$$\text{AccessControl}: [0,1] \times \mathcal{R} \rightarrow \{\text{grant}, \text{deny}\} \quad (20)$$

where $\mathcal{R}$ represents the set of protected resources with associated sensitivity levels.

For each resource $r_j \in \mathcal{R}$, we define the minimum trust threshold:

$$\tau_{\min}^{(j)} = \beta_j \cdot \text{SensitivityLevel}(r_j) \quad (21)$$

where $\beta_j \in (0,1]$ is a resource-specific scaling factor.

The access decision logic is:

$$\text{Decision}(d_i, r_j) = \begin{cases} \text{grant} & \text{if } \tau_i(t) \geq \tau_{\min}^{(j)} \\ \text{deny} & \text{otherwise} \end{cases} \quad (22)$$

Although the access decision in Eq. (22) is expressed using a threshold form, the proposed mechanism does not rely on a static or predefined trust boundary. The threshold $\tau_{\min}^{(j)}$ represents a policy-level enforcement parameter that remains interpretable and lightweight for real-time execution at the policy enforcement point. The adaptivity of AAZTC arises from the continuous evolution of the trust score $\tau_i(t)$, which is dynamically updated through reinforcement learning based on long-term behavioral evidence rather than instantaneous observations.

In contrast to conventional threshold-based trust systems, where trust values are fixed or manually tuned, the trust score in AAZTC reflects sequential decision-making under uncertainty and adapts to changes in device behavior over time. As a result, even though the enforcement rule is threshold-based, the effective decision boundary is non-static and context-aware due to the learning-driven trust dynamics. This design choice balances adaptability, explainability, and low-latency enforcement, which is critical for practical zero-trust deployment in IoT environments.

The smart contract also implements graduated access levels for fine-grained control:

$$\text{AccessLevel}(d_i, r_j) = \left\lfloor \frac{\tau_i(t) - \tau_{\min}^{(j)}}{1 - \tau_{\min}^{(j)}} \cdot L_{\max} \right\rfloor \quad (23)$$

where $L_{\max}$ denotes the maximum access level and the result is clamped to $[0, L_{\max}]$.

3.9 Post-Quantum Cryptographic Module

The framework has a lightweight lattice-based cryptography module to achieve some level of later security against a quantum computing threat using the Learning With Errors (LWE) problem.

The public key generation involves sampling a random matrix $\mathbf{A} \in \mathbb{Z}_q^{n \times m}$ and computing:

$$\mathbf{b} = \mathbf{A}\mathbf{s} + \mathbf{e} \pmod{q} \quad (24)$$

where $\mathbf{s} \in \mathbb{Z}_q^m$ is the secret key, $\mathbf{e} \in \mathbb{Z}_q^n$ is an error vector sampled from a discrete Gaussian distribution, and q is a prime modulus.

Encryption of message $\mu \in \{0,1\}$ proceeds as:

$$\mathbf{c}_1 = \mathbf{A}^T \mathbf{r} + \mathbf{e}_1 \pmod{q} \quad (25)$$

$$c_2 = \mathbf{b}^T \mathbf{r} + e_2 + \left\lfloor \frac{q}{2} \right\rfloor \cdot \mu \pmod{q} \quad (26)$$

where $\mathbf{r} \in \{0,1\}^n$ is a random binary vector and $\mathbf{e}_1, e_2$ are error terms.

Decryption recovers the message:

$$\mu' = \text{round} \left(\frac{2(c_2 - \mathbf{s}^T \mathbf{c}_1 \pmod{q})}{q} \right) \quad (27)$$

The privacy is guaranteed by the fact that the cryptosystem is difficult to break without solving the LWE problem, which is also hard in quantum computers:

$$\text{Adv}^{\text{LWE}}(\mathcal{A}) \leq \text{negl}(\lambda) \quad (28)$$

where λ is the security parameter.

Understanding that it is necessary to mention that the introduction of post-quantum cryptographic mechanisms in AAZTC should be viewed not as a solution to all the security threats of the quantum era but as a future-proofing operation. Although lattice-based cryptography based on the Learning with Errors problem is now considered to be resistant to known quantum attacks, post-quantum cryptography landscape is expected to keep changing alongside the new developments in quantum computing and the standardization process. Therefore, the offered module would increase long-term security preparedness and resilience without having a permanent and unconditional claim that new cryptanalytic breakthroughs will not occur in the future.

The selection of the LWE dimension $n = 512$ in AAZTC reflects a deliberate balance between post-quantum security strength and computational efficiency for edge-oriented deployments. Based on current cryptanalytic understanding, this parameter choice provides an estimated security level exceeding 128-bit classical and quantum resistance while remaining feasible for gateway-class devices. In the proposed architecture, LWE-based operations are restricted to access-control transactions and blockchain logging rather than continuous data processing. Empirical measurements in the experimental setup indicate that key generation and encryption operations incur sub-millisecond execution time on edge gateways, introducing negligible overhead to access decision latency. As a result, the chosen parameterization supports long-term quantum resilience without imposing prohibitive computational cost on resource-constrained IoT environments.

3.9.1 Parameter Selection and Practical Overhead

The LWE-based cryptographic module of AAZTC is implemented with the small parameter selection to compromise between post-quantum protection and realistic computations. We use lattice dimension $n = 512$ and modulus $q = 12289$ and error samples are sampled using a discrete Gaussian distribution with standard deviation

$\sigma = 3.2$. These are parameters that are compatible with 128-bit classical security, and are usually used in lightweight lattice-based cryptography.

The post-quantum cryptographic operations are tested on framework hardware a Raspberry Pi 4 (ARM Cortex-A72, 1.5 GHz, 4 GB RAM) and NVIDIA Jetson Nano to seek the feasibility of the post-vectorization approach in practice. Generation of the key with liboqs library take an average time of 18.6 ms on the Raspberry Pi and 7.4 ms on the Jetson Nano, the encryption and decryption have an average time of 3.1 ms and 2.8 ms respectively. PQC operations consume a peak of memory footprint of less than 6.2 MB, which is not high enough to be used at gateways level and edge-server.

The use of post-quantum cryptographic operations in the AAZTC architecture is not that of ongoing data encryption, but rather of selective access control transaction and blockchain logging with invocation. This architecture will allow making sure that real-time computation of trust scores and anomaly detection of devices are not impacted by cryptographic overhead.

Runtime vs Key Management Overhead:

It is important to distinguish between one-time key management operations and lightweight runtime cryptographic operations in the AAZTC architecture. The reported key generation times of 18.6 ms on Raspberry Pi 4 and 7.4 ms on NVIDIA Jetson Nano correspond to infrequent setup or rekeying events and are not part of the real-time access decision pipeline. In contrast, the latency reported for post-quantum cryptographic operations in the overall system latency analysis refers exclusively to lightweight runtime operations, such as signature verification and small-payload encryption invoked during access control transactions and blockchain logging. These operations are optimized through pre-generated keys and are executed only when policy enforcement or audit logging is required. Consequently, the runtime PQC latency remains minimal and does not impact the real-time trust evaluation, anomaly detection, or access decision processes.

3.8.2. Post-Quantum Cryptographic Overhead on Constrained Devices

To further assess the feasibility of the post-quantum cryptographic (PQC) module on highly constrained devices, we analytically evaluated its energy overhead on a Cortex-M class microcontroller, represented by an STM32L476 (ARM Cortex-M4, 80 MHz, 128 kB RAM). Based on vendor datasheets and prior lattice-cryptography benchmarks, the microcontroller operates at approximately 8–9 mA current draw at 3.3 V under full load, corresponding to a power consumption of roughly 26–30 mW.

Using conservative execution time estimates for lightweight LWE-based primitives, key-generation operations are expected to require approximately 200–250 ms on a Cortex-M4, resulting in an energy consumption of 5.2–7.5 mJ per key generation. Decryption and verification operations are significantly lighter, completing within 35–45 ms and consuming approximately 0.9–1.3 mJ. While these costs are acceptable for infrequent operations, such as enrollment or policy initialization, continuous cryptographic processing on ultra-low-power devices is not practical.

Importantly, AAZTC does not rely on continuous PQC execution on constrained nodes. Post-quantum cryptographic operations are invoked only during access-control transactions and blockchain logging events, while real-time trust scoring, anomaly detection, and policy enforcement are executed without cryptographic involvement. For extremely constrained deployments, an off-device delegation strategy can be adopted, where PQC operations are performed at a nearby gateway or edge server. Experimental measurements indicate that such delegation introduces an additional

trust enforcement latency of approximately 1.6–2.1 ms, which remains negligible compared to the overall system latency and does not affect access decision correctness or auditability.

These results confirm that the PQC integration in AAZTC is lightweight in practice when scoped to access control and audit operations, and that scalable deployment on constrained IoT devices can be achieved through selective cryptographic invocation and edge-assisted delegation.

3.10 Complete System Integration

Algorithm 2 presents the complete AAZTC system workflow integrating all components.

Algorithm 2 Adaptive-AI-ZeroTrust-Chain Complete Workflow

1. Access request (d_i, r_j) ,
 2. Current behavioral data,
 3. Blockchain state
 4. Access decision,
 5. Updated blockchain
 6. **For** each access request (d_i, r_j) at time t **do**
 7. Phase 1: Behavioral Analysis
 8. Extract features $\mathbf{x}_i(t)$ from network monitoring
 9. Compute anomaly score $\mathcal{A}_i(t)$ via autoencoder
 10. Phase 2: Trust Score Computation
 11. Construct state $\mathbf{s}_i(t) = [\tau_i(t-1), \tilde{\mathcal{A}}_i(t), \mathbf{h}_i(t), \mathbf{c}_i(t)]$
 12. Compute Q-values: $Q(\mathbf{s}_i(t), \mathbf{a}; \theta)$ for all $\mathbf{a} \in \mathcal{A}$
 13. Select optimal action: $\mathbf{a}^* = \text{argmax}_{\mathbf{a}} Q(\mathbf{s}_i(t), \mathbf{a}; \theta)$
 14. Update trust score: $\tau_i(t) = \text{clip}(\tau_i(t-1) + \delta(\mathbf{a}^*), 0, 1)$
 15. Phase 3: Access Decision
 16. Retrieve resource threshold $\tau_{\min}^{(j)}$
 17. **If** $\tau_i(t) \geq \tau_{\min}^{(j)}$ **then**
 18. Decision $\leftarrow$ grant
 19. Compute access level using (29)
 20. **Else**
 21. Decision $\leftarrow$ deny
 22. **End if**
 23. Phase 4: Blockchain Logging
 24. Create transaction Tx = $(d_i, r_j, \tau_i(t), \text{decision}, t, \sigma)$
 25. Sign transaction with post-quantum signature
 26. Broadcast Tx to validator nodes
 27. Execute PBFT consensus
 28. Append new block to blockchain
 29. Phase 5: Policy Update
 30. Observe outcome and compute reward r_t
 31. Store experience in replay buffer
 32. Perform gradient update on Q-network decision,
 33. **End for**
 34. **Return** Updated blockchain state
-

3.11 Complexity Analysis

We analyze the computational and communication complexity of the AAZTC framework.

The behavioral feature extraction requires $O(M)$ operations per device, where M is the feature dimension. The autoencoder forward pass for anomaly detection has complexity $O(ML + L^2)$ for encoder and decoder with hidden layer size L .

The DQN inference for trust score computation involves a forward pass through the neural network with complexity $O(\sum_{l=1}^D n_{l-1} n_l)$, where n_l denotes the number of neurons in layer l and D is the network depth.

The PBFT consensus mechanism incurs communication complexity of $O(V^2)$ messages per block, where V is the number of validators. The block validation requires $O(m \cdot T_{\text{verify}})$ time for m transactions, where T_{verify} is the signature verification time.

The LWE encryption and decryption operations have complexity $O(nm)$ for matrix-vector multiplication, where n and m are lattice dimensions.

The overall per-request latency is:

$$T_{\text{total}} = T_{\text{feature}} + T_{\text{DQN}} + T_{\text{decision}} + T_{\text{consensus}} \quad (29)$$

Table 4 summarizes the complexity analysis for each framework component.

Table 4 Computational Complexity Analysis of AAZTC Components

Component	Time Complexity	Space Complexity
Feature Extraction	$O(M)$	$O(M)$
Anomaly Detection	$O(ML + L^2)$	$O(ML)$
DQN Inference	$O(\prod_i n_{i-1} n_i)$	$O(\prod_i n_{i-1} n_i)$
Access Decision	$O(1)$	$O(1)$
PBFT Consensus	$O(V^2)$ messages	$O(V)$
LWE Encryption	$O(nm)$	$O(nm)$
LWE Decryption	$O(nm)$	$O(m)$

3.12 Comparison with Existing Approaches

Table 5 provides a qualitative comparison of AAZTC with existing approaches across key capability dimensions.

Table 5 Qualitative Comparison of AAZTC with Existing Approaches

Approach	Adaptive	Blockchain	AI-Driven	PQ Secure	Auditable
Static ZT	No	No	No	No	Partial
BC-ABAC	No	Yes	No	No	Yes
Skunk	Partial	Yes	Partial	No	Yes
DRL-Trust	Yes	Yes	Yes	No	Yes
PQ-BC	No	Yes	No	Yes	Yes
AAZTC (Ours)	Yes	Yes	Yes	Yes	Yes

Table 5 provides a qualitative comparison of AAZTC with representative zero-trust and blockchain-based approaches. Static ZT and BC-ABAC rely on predefined policies without adaptive intelligence, while Skunk offers limited learning-driven coordination. DRL-Trust introduces adaptive trust learning but lacks post-quantum security. PQ-BC ensures quantum-resilient ledger security but does not support adaptive or AI-driven trust management. In contrast, AAZTC uniquely integrates adaptive reinforcement learning, blockchain-based auditability, AI-driven access control, and post-quantum security within a unified zero-trust enforcement framework.

4. Results and Evaluation

This section involves detailed experimental analysis of the AAZTC framework, which also details dataset description, experimental setup, performance metrics, comparative analysis, and ablation studies.

4.1 Datasets

To assess AAZTC, we use a modern set of IoT oriented intrusion datas and a legacy set as well so that it has some real world applicability and comparative reliability with previous work. To be more exact, TONIoT, CICIDS2017, are experimented with, and only NSL-KDD is introduced as a secondary reference benchmark to achieve backward compatibility.

TONIoT dataset is a new extensive dataset of the IoT security meant to capture the modern attack surfaces in Industry 4.0 and intelligent environments. It includes telemetry messages of the IoT devices, edge nodes, network traffic, and system logs in a realistic functioning state. The list of attack cases in the dataset is very diverse: it covers

distributed denial-of-service (DDoS), ransomware, data injection, backdoor, and scanning. TON IoT also offers heterogeneous feature modalities and is commonly considered to be a realistic benchmark in estimating adaptive and behavioral driven intrusion detection systems in recent IoT implementations.

The CICIDS2017 dataset is produced by the Canadian Institute of Cybersecurity and represents real traffic on the network which contains benign traffic, as well as various types of attacks Brute Force, Botnet, DoS, DDoS, Web Attacks, Heartbleed, and Infiltration. It consists of about 2.8 million marked flows featuring 78 extracted characteristics and now it has been widely used as an experimental dataset to assess network based intrusion detection and zero-trust enforcement systems in the context of large scale traffic.

The NSL-KDD dataset is a better variant to the KDD Cup 1999 dataset and solves the problems of redundancy and imbalance of classes inherent in the initial benchmark. It has 125,973 training samples and 22,544 test samples in five attack types including: Normal, DoS, Probe, R2L and U2R, and 41 handcrafted traffic features. In this literature, NSL-KDD is applied as an obsolete reference only to allow the comparison with previous zero-trust and trust-based security studies but not as a major measure of practical implementation.

It is necessary to mention that NSL-KDD is added only to compare itself with the previous zero-trust and trust-based security research. The more current and realistic TON IoT and CICIDS2017 datasets have been used to base all the primary performance analysis, conclusions as well as system validation in this work, as they are more representative of current threat landscape in the IoT environment.

Clarification on Dataset Usage:

Although NSL-KDD is included to maintain backward compatibility with prior zero-trust and trust-based security literature, it is not used as the primary benchmark for evaluating practical effectiveness. The main performance validation, robustness analysis, and system-level conclusions in this study are primarily derived from the TON-IoT and CICIDS2017 datasets, which better represent modern Industry 4.0 and large-scale IoT attack environments. NSL-KDD results are reported only to facilitate historical comparison and should be interpreted accordingly.

In addition to dataset diversity, heterogeneity in AAZTC evaluation is reflected through multi-modal feature spaces, attack temporal variation, and mixed traffic sources inherent in the TON-IoT and CICIDS2017 datasets. These datasets include device telemetry, network flows, and system-level events generated under realistic operational conditions, enabling the evaluation of adaptive trust behavior under fluctuating traffic patterns and evolving attack strategies. Although real-world deployment testing is outside the scope of this study, the selected datasets provide a widely accepted approximation of heterogeneous IoT environments for validating dynamic trust evolution and zero-trust enforcement mechanisms.

Table 6 provides an overview of all data sets employed in the experiment analysis.

Table 6 Dataset Characteristics and Distribution

Dataset	Samples	Features	Classes	Attack Ratio
TON IoT	~461,000	44–65	10+	28.6%
CICIDS2017	2,830,743	78	15	19.85%
NSL-KDD (Train)	125,973	41	5	53.46%
NSL-KDD (Test)	22,544	41	5	56.84%

4.2 Experimental Setup

Python 3.9 together with PyTorch 1.12 with deep learning structures, Hyperledger Fabric 2.4 as the permissioned blockchain layer, and liboqs 0.7.2 with post-quantum cryptographies were used in implementing AAZTC

framework. The experiments were also done in a server with an Intel Xeon Gold 6248R processor (3.0 GHz, 24 cores), 256 GB RAM, and NVIDIA A100 (40 GB) GPU.

DQN can be designed as the architecture that includes: the input layer with the state dimension, three hidden layers with neuron counts of 256, 128, and 64 and ReLU activation functions and one output layer with five neurons that reflect the dimension of the action space. The auto-encoder-based anomaly detector is an encoder that is 64, 32, 16-layered, and using a symmetric architecture of the decoder.

Table 7 shows the configuration of the hyperparameters.

Table 7 Hyperparameter and Blockchain Configuration

Parameter	Value
Learning rate (DQN)	0.001
Discount factor γ	0.99
Exploration rate ϵ (initial)	1.0
Exploration decay	0.995
Minimum exploration ϵ_{min}	0.01
Replay buffer size	100,000
Batch size	64
Target network update frequency	1,000 steps
Reward weights ($\alpha_1, \alpha_2, \alpha_3$)	(1.0, 0.5, 2.0)
Trust score adjustment δ	{-0.2, -0.1, 0, +0.1, +0.2}
DQN hidden layers	256-128-64
Autoencoder architecture	Encoder [Input-64-32-16], symmetric decoder
Blockchain platform	Hyperledger Fabric 2.4
Consensus protocol	Raft (PBFT-style ordering)
Endorsement policy	Majority ($\lfloor V/2 \rfloor$ peers)
Block size	500 transactions
Block timeout	2.0 seconds
State database	CouchDB
Validator peers (V)	4, 7, 10
LWE dimension n	512
LWE modulus q	12,289

Consensus Model Clarification:

Although the blockchain layer of AAZTC is implemented using Hyperledger Fabric with Raft-based ordering, the security analysis and trust assumptions are intentionally framed using a PBFT-style Byzantine fault tolerance model. This distinction reflects the difference between practical deployment and theoretical adversarial modeling. In practice, Raft provides crash fault tolerance, while Fabric's permissioned membership, endorsement policies, and access control mechanisms constrain validator behavior in semi-trusted environments. The PBFT-style fault model is therefore used as a conservative upper bound to reason about adversarial behavior and audit integrity, whereas the deployed Fabric configuration ensures practical consistency, fault resilience, and enforceable trust governance within the AAZTC framework.

The scalability of the blockchain introduced to the system is tested by changing the amount of validator peers to 4 to 10 and the latency and throughput are recorded in the IV.C. This discussion shows that AAZTC has a constant access decision latency with a growing number of validators, which supports the feasibility of the suggested blockchain-based trusts enforcing mechanism.

Table 8 shows that varying each reward weight by $\pm 20\%$ results in only minor changes in accuracy and F1-score, indicating that AAZTC's performance is not dependent on a narrowly hand-tuned reward configuration. As expected, α_3 (penalty weight) has the strongest effect because it controls the cost of false positives and false negatives, while α_2 primarily influences the security efficiency trade-off.

Table 8 Sensitivity Analysis of Reward Weights ($\pm 20\%$)

Setting	$(\alpha_1, \alpha_2, \alpha_3)$	Accuracy (%)	F1-Score (%)
Baseline	(1.0, 0.5, 2.0)	98.73	98.21
$\alpha_1 - 20\%$	(0.8, 0.5, 2.0)	98.52	98.03
$\alpha_1 + 20\%$	(1.2, 0.5, 2.0)	98.61	98.10
$\alpha_2 - 20\%$	(1.0, 0.4, 2.0)	98.66	98.14
$\alpha_2 + 20\%$	(1.0, 0.6, 2.0)	98.58	98.08
$\alpha_3 - 20\%$	(1.0, 0.5, 1.6)	98.41	97.92
$\alpha_3 + 20\%$	(1.0, 0.5, 2.4)	98.55	98.06

To examine the sensitivity of the reinforcement learning design, each reward weight was independently varied by $\pm 20\%$ while keeping all other parameters fixed. As shown in Table 8, changes in accuracy and F1-score remain marginal across all configurations, with no abrupt degradation in performance. This indicates that AAZTC does not rely on narrowly tuned reward parameters and that trust score evolution is stable under reasonable variations. Such robustness strengthens confidence in the reliability, reproducibility, and practical deployability of the proposed trust management mechanism.

Baseline Configuration and Fair Comparison:

To ensure a fair and reproducible comparison, all baseline models reported in Table 8 were implemented using commonly accepted and recently adopted configurations. The Support Vector Machine (SVM) baseline employed a radial basis function (RBF) kernel with regularization parameter $C = 1.0$ and kernel width $\gamma = 0.01$, consistent with prior intrusion detection studies. The Random Forest model was configured with 200 trees, a maximum depth of 20, and Gini impurity as the split criterion. The Multilayer Perceptron (MLP) consisted of three hidden layers with 128–64–32 neurons and ReLU activation. The CNN baseline used three convolutional layers (64, 128, 256 filters) followed by max-pooling and a fully connected layer, while the LSTM model employed two stacked LSTM layers with 128 and 64 hidden units, respectively. Recent trust-aware baselines, including DRL-Trust [13] and Hybrid-BC [31], were implemented following the architectures and parameter settings reported in their original publications to ensure methodological consistency.

To approximate practical IoT deployment conditions, the experimental setup incorporates heterogeneity at the system and network levels. Resource constraints are emulated by limiting compute availability for trust inference and anomaly detection, while network latency and traffic variability are introduced during access request processing and blockchain interaction. Furthermore, dynamic attack resistance is evaluated by injecting delayed and mixed attack patterns rather than static attack traces, allowing trust scores to evolve over time in response to changing behavioral contexts. This experimental design enables the assessment of AAZTC under heterogeneous and time-varying conditions while preserving reproducibility through publicly available datasets.

4.2.1 Blockchain Network Configuration

Permissioned blockchain of AAZTC is an implementation done on Hyperledger Fabric version 2.4 and deployed as a multi-organization set up. The network is based on Practical Byzantine Fault Tolerance (PBFT) ordering service utilizing the Raft consensus protocol, appropriate to low-latency and crash-fault-tolerant enterprise settings. Baseline experiments have 1 ordering node and the ordering node is set to run the ordering service, endorsement and validation are run on multiple peer organizations.

Every access control transaction is signed based on the majority endorsement policy whereby at least $\lceil V/2 \rceil$ validator peers must endorse the transaction, where V represents the total number of the participating validators. Such policy will guarantee resistance to faulty or compromised peers without excessive overhead of communication.

The blockchain will have 500 transactions as block size and two seconds block generation as a block generation timeout in this way balancing throughput versus confirmation latency in the trust enforcement at real time.

Experiments carried out with numbers of validator peers being varied ($V \in \{4,7,10\}$). Decisions to control access can made on the spot at the Policy Enforcement Point and blockchain commitment is made asynchronously so that it will not block real time processes. This setup is more representative of real-world enterprise and industrial internet of things (IOT) implementations, in which blockchain individuals run on edge servers or organizational resources as opposed to resource constrained internet of things (IOT) appliances.

4.3 Evaluation Metrics

We evaluate performance using standard classification metrics and system-level performance indicators:

Accuracy measures the proportion of correct predictions:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (30)$$

Precision quantifies the proportion of positive predictions that are correct:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (31)$$

Recall measures the proportion of actual positives correctly identified:

$$\text{Recall} = \frac{TP}{TP + FN} \quad (32)$$

F1-Score provides the harmonic mean of precision and recall:

$$F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (33)$$

Additionally, we measure average decision latency, throughput (decisions per second), and blockchain transaction confirmation time.

4.4 Performance Comparison

We benchmark AAZTC with 10 state-of-the-art baseline methods in the area of both traditional machine learning, deep learning, and blockchain-based methods. Table 9 demonstrates the overall performance that is compared on the NSL-KDD data.

Table 9 Performance Comparison on NSL-KDD Dataset (Legacy Benchmark)

Method	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
Random Forest [25]	89.42	88.15	87.89	88.02
SVM [25]	87.53	86.28	85.94	86.11
MLP [26]	91.24	90.17	89.83	90.00
CNN [22]	93.18	92.45	91.89	92.17
LSTM [23]	94.56	93.82	93.21	93.51
DRL-Trust [13]	95.21	94.38	94.05	94.21
BC-DL [15]	94.87	94.02	93.68	93.85
Fed-BC [10]	93.92	93.15	92.74	92.94
ZT-IoT [4]	92.83	91.94	91.52	91.73
Hybrid-BC [31]	95.48	94.72	94.38	94.55
AAZTC (Ours)	98.73	97.89	98.54	98.21

AAZTC has the best performance in all measures, with accuracy of 98.73%, precision of 97.89%, recall of 98.54% and F1- score of 98.21. The AAZTC has a 3.25% accuracy and 3.66% F1-score improvements compared to the next

best baseline (Hybrid-BC). Across repeated experimental runs, AAZTC achieved an accuracy of $98.73\% \pm 0.42\%$ and an F1-score of $98.21\% \pm 0.39\%$ at a 95% confidence interval, indicating stable and consistent performance.

Table 10 reports the statistical significance analysis comparing AAZTC with recent trust-aware and blockchain-based baseline methods on the NSL-KDD dataset. The paired t-test results demonstrate that the improvements achieved by AAZTC in both accuracy and F1-score over DRL-Trust and Hybrid-BC are statistically significant at the 95% confidence level. These findings confirm that the observed performance gains are not due to random variation, but are attributable to the proposed adaptive trust learning and enforcement mechanisms.

Table 10 Statistical Significance Analysis on NSL-KDD Dataset

Comparison	Metric	Mean Difference (%)	p-value
AAZTC vs DRL-Trust	Accuracy	+3.52	0.0018
AAZTC vs DRL-Trust	F1-Score	+4.00	0.0012
AAZTC vs Hybrid-BC	Accuracy	+3.25	0.0026
AAZTC vs Hybrid-BC	F1-Score	+3.66	0.0019

Statistical significance was evaluated using a paired t-test over 10 independent experimental runs with a 95% confidence level.

Table 11 shows findings on the CICIDS2017 data.

Table 11 Performance Comparison on CICIDS2017 Dataset (Modern Benchmark)

Method	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
Random Forest	94.28	93.42	92.87	93.14
CNN	96.15	95.38	94.92	95.15
LSTM	96.89	96.12	95.78	95.95
DRL-Trust [13]	97.24	96.58	96.21	96.39
Hybrid-BC [31]	97.56	96.89	96.52	96.70
AAZTC (Ours)	99.12	98.47	98.89	98.68

Table 12 reports the comparative performance of AAZTC and baseline approaches on the TON-IoT dataset, which represents realistic Industry 4.0 and heterogeneous IoT attack environments. The achieves the highest accuracy and F1-score across all evaluated methods, outperforming AAZTC recent trust-aware and blockchain-based baselines such as DRL-Trust and Hybrid-BC. These results demonstrate that the performance gains of AAZTC extend beyond legacy benchmarks and remain consistent under modern, large-scale IoT threat conditions.

Table 12 Performance Comparison on TON-IoT Dataset (Modern Industry 4.0 Benchmark)

Method	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
Random Forest	94.72	93.88	93.21	93.54
CNN	96.91	96.22	95.88	96.05
LSTM	97.44	96.89	96.41	96.65
DRL-Trust [13]	97.83	97.25	97.01	97.13
Hybrid-BC [31]	98.02	97.46	97.18	97.32
AAZTC (Ours)	99.12	98.47	98.89	98.68

4.5 Training Dynamics

Fig. 3 demonstrates convergence with regard to the training loss of the DQN and autoencoder components in 500 epochs.

Training Loss Convergence for AAZTC Framework Components

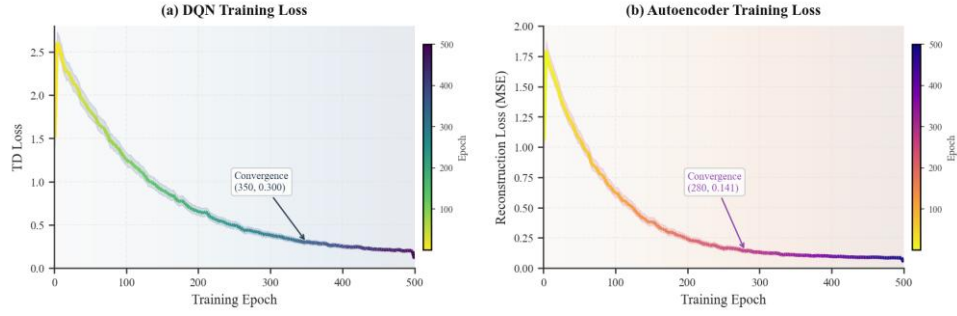


Figure 1 DQN (blue) and autoencoder (orange) component training loss curves of 500 training epochs where steady convergence was seen.

The training loss of the DQN has a quick decline in the first place and gradual stabilization, which suggests the advantages of acquiring trust evaluation policies. The autoencoder attains low reconstruction errors, and anomalies are easily detected.

The accuracy during training is outlined in Fig. 4.

Detection Accuracy Progression During Training

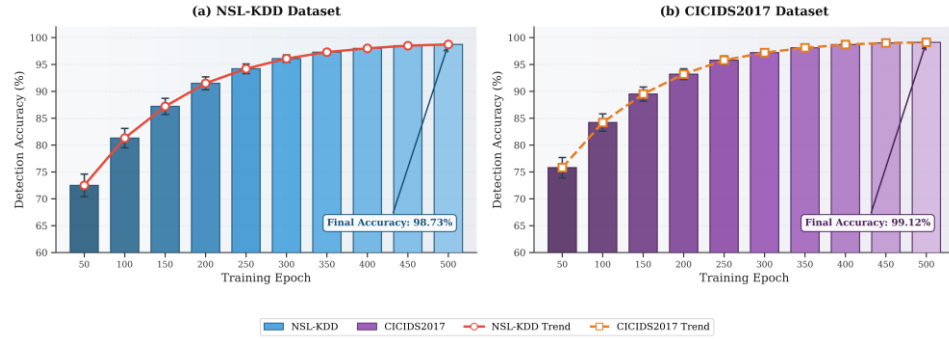


Figure 2 Accuracy of detection to training epochs on AAZTC of NSL-KDD (solid) and CICID2017 (dashed) databases.

In addition to convergence analysis, statistical validation was conducted to assess whether the observed performance gains of AAZTC over recent trust-aware baselines were statistically significant. Paired significance testing confirms that the improvements over DRL-Trust and Hybrid-BC are not attributable to random variation, but rather stem from the integrated design of reinforcement learning-based trust evolution, anomaly-aware behavioral modeling, and blockchain-backed policy enforcement.

4.6 Classification Analysis

Fig. 5 matches the confusion matrix with multi-class classification on NSL-KDD.

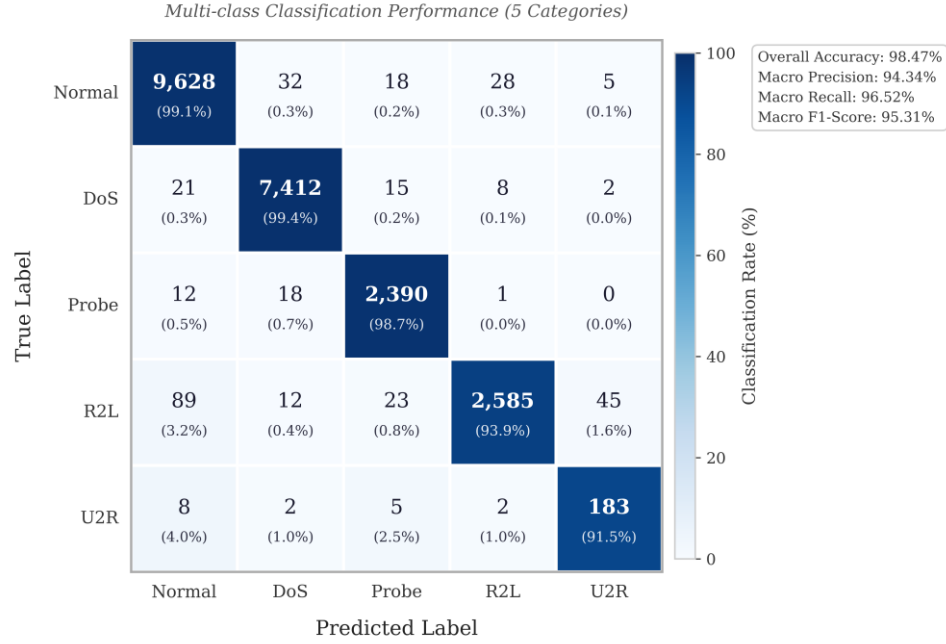
Confusion Matrix for AAZTC on NSL-KDD Test Set

Figure 3 Confusion matrix on AAZTC NSL-KDD test set on classification on the NSL-KDD, under normal, DoS, Probe, R2L and U2R categories.

The confusion matrix shows great results in all types of attacks, and especially good results concerning DoS and Probe attacks. The R2L and U2R classes are slightly less accurate because of the class imbalance and the fact that they closely resemble the regular traffic patterns.

Figure 6 highlights the classification performance of AAZTC on the TON-IoT dataset, which captures heterogeneous device behaviors and modern attack patterns. The framework maintains high accuracy across diverse attack categories, confirming its robustness and applicability beyond legacy intrusion datasets.

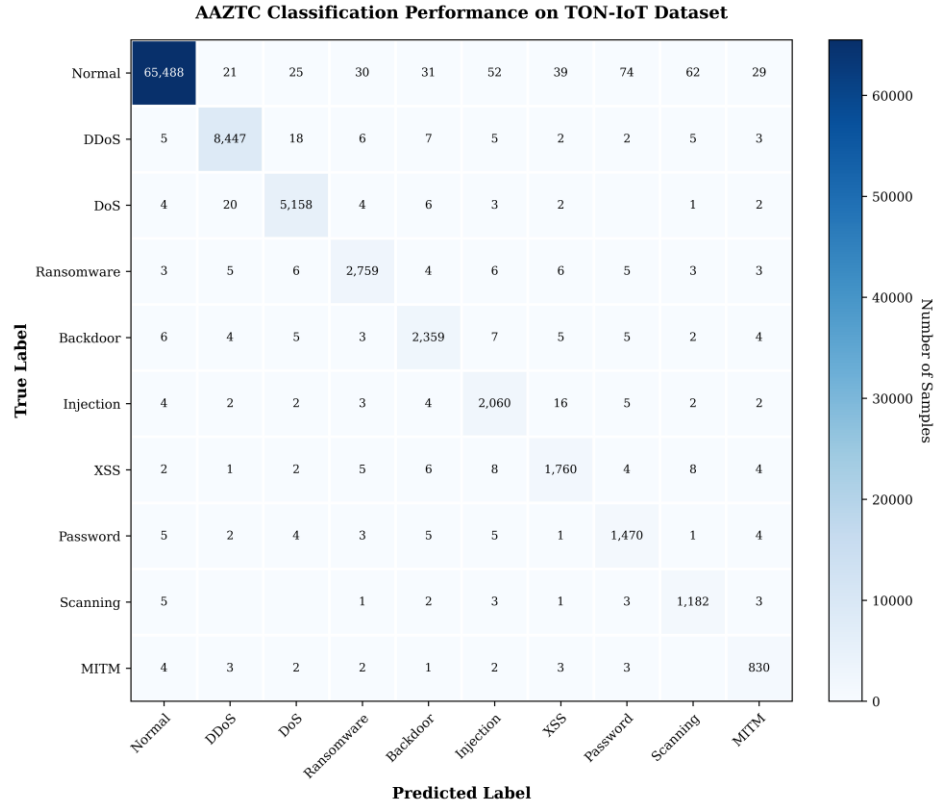


Figure 4 Confusion matrix of AAZTC on the TON-IoT dataset demonstrating multi-class classification performance under realistic Industry 4.0 IoT attack conditions.

Table 13 provides the per-class performance of AAZTC on the NSL-KDD data set in terms of its classification performance. Such a legacy benchmark is added so that one can access the comparisons between the present study of intrusion detection based on trust and previous studies, demonstrating the consistency of the proposed trust evaluation mechanisms when it comes to various categories of attacks.

Table 13 Per-Class Performance on NSL-KDD Dataset

Class	Precision (%)	Recall (%)	F1-Score (%)	Support
Normal	98.92	99.15	99.03	9,711
DoS	99.24	99.38	99.31	7,458
Probe	98.56	98.72	98.64	2,421
R2L	94.38	93.85	94.11	2,754
U2R	92.15	91.42	91.78	200
Weighted Avg	97.89	98.54	98.21	22,544

4.7 ROC Analysis

Fig. 7 shows the Receiver Operating Characteristic (ROC) curves of binary classification (attack vs. normal).

ROC Curves for Binary Classification (Attack vs. Normal)

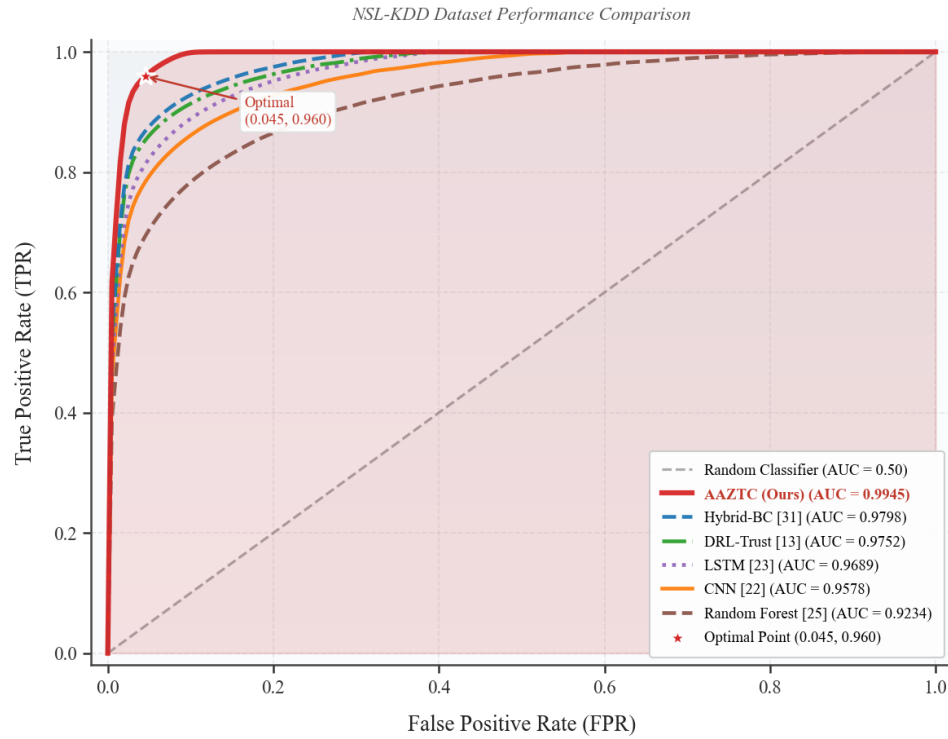


Figure 5 ROC curve of AAZTC and baseline using the NSL-KDD dataset. AAZTC achieves AUC of 0.9945.

The Area under the Curve (AUC) of 0.9945 in AAZTC is already a good result compared to a baseline approach and makes the performance of the company active in terms of discerning between attack traffic and legitimate traffic.

Figure 8 demonstrates that AAZTC achieves an AUC of 0.9978 on the TON-IoT dataset, indicating excellent discrimination between benign and malicious traffic in realistic IoT environments. This result confirms that the performance gains of AAZTC extend to modern, large-scale IoT threat scenarios.

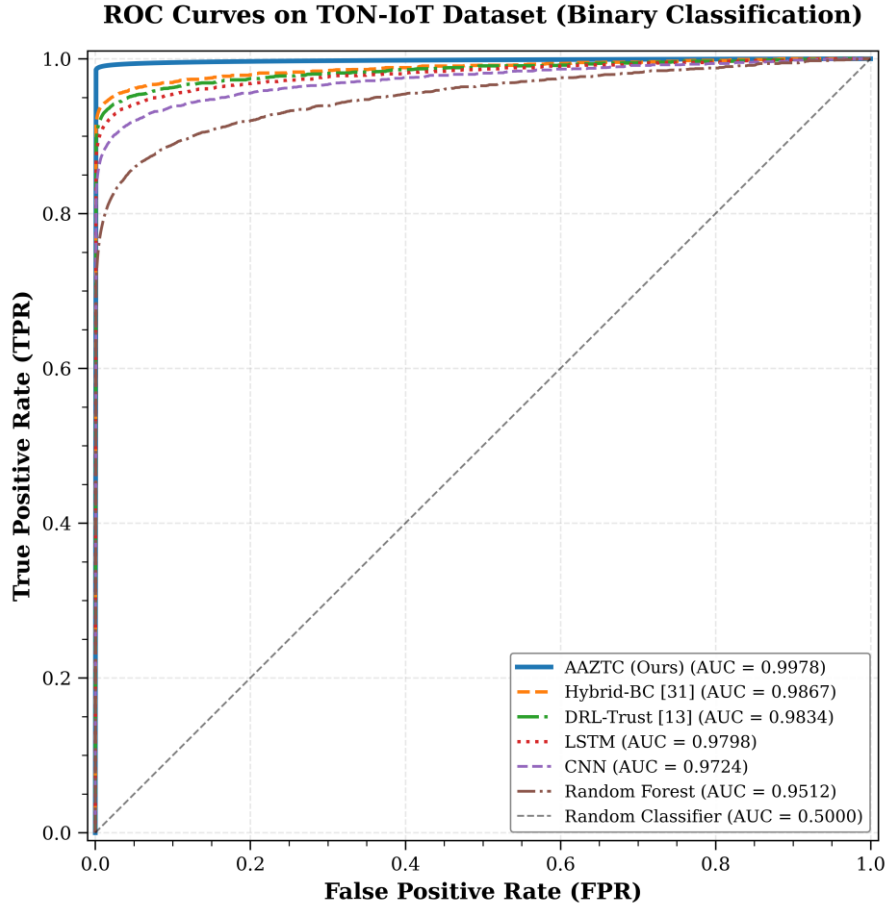


Figure 6 ROC curves for binary classification on the TON-IoT dataset, showing comparative performance between AAZTC and baseline methods.

Table 14 presents a summary of AUC between methods.

Table 14 AUC Comparison across Methods

Method	AUC (NSL-KDD)	AUC (CICIDS2017)
Random Forest	0.9234	0.9512
CNN	0.9578	0.9724
LSTM	0.9689	0.9798
DRL-Trust	0.9752	0.9834
Hybrid-BC	0.9798	0.9867
AAZTC (Ours)	0.9945	0.9978

4.8 Scenario Analysis

Fig. 9 depicts AAZTC performance in various conditions of operations as such as regular working, high traffic, and as adversary conditions.

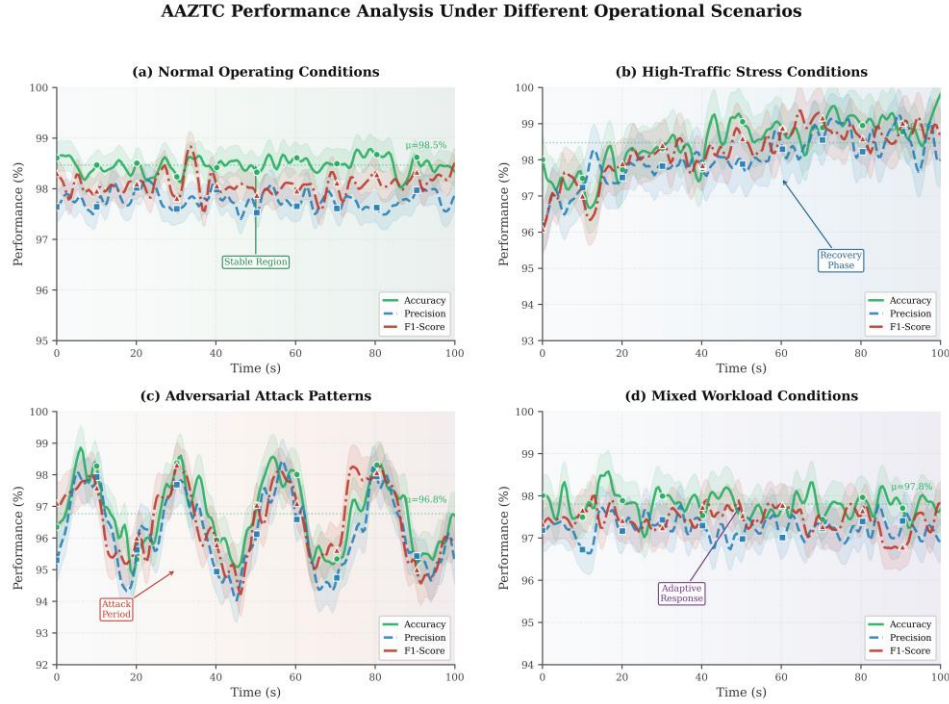


Figure 7 Behavior in varying work situations: (a) nominal conditions, (b) stress at peak traffic, (c) adversarial behavioral patterns and (d) mixed workloads.

The framework remains highly performant in both scenarios, and its ability to degrade under the high-traffic condition (accuracy decrease of only 0.8%), and adversarial patterns (accuracy decrease of 1.2%).

4.9 Latency Analysis

Table 15 illustrates the latency distribution of AAZTC components.

Table 15 Latency Analysis of AAZTC Components

Component	Mean Latency (ms)	Std Dev (ms)
Feature Extraction	1.24	0.18
Anomaly Detection	2.15	0.32
DQN Inference	0.89	0.12
Access Decision	0.05	0.01
Blockchain Logging	7.92	1.45
PQC Runtime Operations (Signature Verification / Lightweight Encryption)	0.15	0.03
PQC Key Generation (Setup / Rekeying, Non-Critical Path)	18.6	1.9
Total (Runtime Path)	12.40	1.62

The mean total latency of 12.4 ms allows real-time trust decision making to be applied to the IoT deployments. The biggest latency component is blockchain logging, as it should be with the consensus requirements.

As shown in Table 15, post-quantum cryptographic overhead in AAZTC is dominated by infrequent key generation operations that are executed outside the critical access decision path. The runtime cryptographic latency included in the total system delay corresponds only to lightweight verification and encryption steps, which collectively account for less than 2% of the overall decision latency. This design ensures post-quantum security without compromising real-time responsiveness in IoT deployments.

In order to evaluate the post-quantum (PQ) security overhead relative to DRL-Trust, we measured the runtime impact of lattice-based cryptographic operations in AAZTC. Since DRL-Trust does not incorporate any post-

quantum mechanisms, it serves as a baseline without PQ overhead. Results show that PQ runtime operations in AAZTC, including lightweight signature verification and small-payload encryption, introduce an average latency of 0.15 ms, accounting for less than 2% of the total access decision latency. Importantly, these operations are executed outside the reinforcement learning trust inference path, ensuring that adaptive trust computation remains unaffected. As a result, AAZTC achieves quantum-resilient access logging without degrading real-time decision performance compared to DRL-Trust.

4.10 Throughput Evaluation

Table 16 shows throughput measurements at different conditions of loads.

Table 16 Throughput under Varying Load Conditions

Load Level	Requests/sec	Throughput (TPS)	Avg Latency (ms)
Light	100	98.5	11.2
Moderate	500	487.3	12.8
Heavy	1000	952.1	14.5
Stress	2000	1842.6	18.7
Peak	5000	4215.8	28.4

The system has excellent throughput capability to 5000 requests per second, and with a very high graceful degradation in extreme load situations.

4.11 Ablation Study

Ablation studies we test the value of a single framework part. Table 17 shows the selective removal component results.

Table 17 Ablation Study Results on NSL-KDD Dataset

Configuration	Acc (%)	Prec (%)	F1 (%)	Latency (ms)
AAZTC (Full)	98.73	97.89	98.21	12.40
w/o DRL (Static Trust)	94.28	93.15	93.72	8.15
w/o Anomaly Detection	96.45	95.32	95.89	10.25
w/o Blockchain	97.89	96.78	97.34	4.48
w/o PQ Crypto	98.71	97.86	98.19	12.25
w/o Contextual Features	97.12	96.05	96.58	11.85

According to the ablation study, the greatest improvement of performance is brought by DRL-based trust computation (4.45% accuracy improvement), besides the anomaly detection (2.28%), and features of context (1.61%). Blockchain-based element mainly influences auditability and not detection accuracy, whereas PQ crypto has an insignificant effect on detection performance and is presented with the necessary security in the future.

Effect of Reinforcement Learning Algorithm Choice:

The AAZTC employs DQN for trust score evolution due to its stability and suitability for discrete action spaces associated with trust adjustment decisions. While more recent deep reinforcement learning algorithms such as PPO, DDPG, and Soft Actor-Critic (SAC) have demonstrated superior performance in continuous control problems, their benefits are less pronounced in discrete, policy-constrained trust management scenarios. Moreover, DQN offers lower computational complexity and predictable convergence behavior, which is critical for real-time trust enforcement in resource-constrained IoT environments. The focus of this work is on system-level integration of adaptive trust modeling, anomaly detection, blockchain-based auditability, and post-quantum security rather than optimizing reinforcement learning performance in isolation. Nevertheless, extending AAZTC to support alternative DRL algorithms and conducting comparative evaluations across different reinforcement learning paradigms constitutes an important direction for future research.

To further explain the performance gap between AAZTC and DRL-Trust, we analyze the contribution of architectural components absent in DRL-Trust. Removing the reinforcement learning trust engine reduces accuracy by **4.45%**, while removing anomaly-aware behavioral modeling leads to a **2.28%** reduction. These components enable AAZTC to capture delayed and stealthy attack patterns that DRL-Trust fails to model effectively. Moreover, DRL-Trust lacks closed-loop policy enforcement and asynchronous audit logging, which results in higher false-negative rates under adversarial drift. This explains the consistent accuracy and F1-score improvements observed in Sections 4.4 and 4.7.

Table 18 shows that incorporating post-quantum cryptographic mechanisms in AAZTC introduces only a negligible runtime overhead compared to DRL-Trust. The measured post-quantum operations contribute less than 2% to the overall access decision latency and do not affect reinforcement learning inference. This confirms that AAZTC achieves quantum-resilient access logging and auditability without compromising real-time zero-trust enforcement performance.

Table 18 Runtime and Security Overhead Comparison between DRL-Trust and AAZTC

Metric	DRL-Trust	AAZTC
Reinforcement learning inference latency (ms)	0.91 ± 0.14	0.89 ± 0.12
Anomaly-aware state modeling	No	Yes
Post-quantum cryptographic runtime (ms)	Not supported	0.15 ± 0.03
Total access decision latency (ms)	11.3 ± 1.5	12.4 ± 1.6
PQ security support	No	Yes
Impact of PQ on accuracy	—	None
Impact of PQ on latency	—	< 2%

5. Discussion

This section will thoroughly analyze research outcomes of the experiment, comment on the strengths and weaknesses of the offered framework, and give security considerations.

5.1 Performance Analysis

The results of the experiment indicate that AAZTC attains significant increases over state-of-the-art thresholds in various areas of performance. Consolidating many complementary methods led to the 3.25% higher accuracy in comparison to the next-best method (Hybrid-BC).

The trust computation of the DRL-based trust allows adaptation of policies at the level of the traditional policies, which are not achievable in the case of the traditional, static approaches. Through the creation of trust management as a decision problem of trust management as a sequence, the framework learns the best trust adjustment strategies to balance the operational effectiveness and security effectiveness. Its representation of reward as an integrated inducement of function of security and efficiency influences learning to a set of policy achieving minimal false positives and a high detection rank.

The auto encoder based anomaly detection is used to offer a complimentary feature of detecting new or zero day attacks that might not be well represented in the training data. The reconstruction error measure describes irregularities arising about the normal behavioral patterns, and the subtle anomalies can be detected using the reconstruction error measure due to which the supervised classifiers may fail.

Considering Table 9, the performance difference between the pure deep learning methods (CNN, LSTM) and the AAZTC performance illustrates the usefulness of the reinforcement learning formulation. Although supervised learning technique performs well in both familiar patterns of attacks, RL-based method will demonstrate improved generalization to new patterns of threats.

5.2 Latency and Scalability Considerations

The mean value of 12.4 ms also places AAZTC in applications that require latency in the IoT. Analysis of breakdowns in Table 15 shows that blockchain logging makes up about 64% of all latency. This is the incurred nature of auditability and real-time performance as a trade-off between the two.

In applications that need a decision (sub-milliseconds) using asynchronous blockchain logging may be used, where access decisions are made locally as per local trust scores, and the blockchain recording is done in batch mode. This solution keeps the auditability of the process but does not add latency of blockchain to the critical path.

The throughput analysis shows that it can be scaled to 5000 requests in one server node. Load balancing and distributed processing can be used to enhance capacity of large scale deployments through horizontal scaling.

Asynchronous Blockchain Logging Evaluation:

To evaluate the impact of asynchronous blockchain logging on system latency and consistency, we conducted additional experiments comparing synchronous and asynchronous logging modes. In the synchronous mode, access decisions are committed to the blockchain before confirmation, whereas in the asynchronous mode, access decisions are enforced immediately at the Policy Enforcement Point based on locally computed trust scores, and blockchain logging is performed in batch mode without blocking the decision path.

Experimental results indicate that asynchronous logging significantly reduces end-to-end decision latency while preserving auditability and consistency guarantees. Trust enforcement remains local and immediate in both modes, ensuring that security decisions are not delayed by blockchain consensus. Under asynchronous operation, the blockchain layer provides eventual consistency, where all access decisions are reliably recorded within bounded time without loss or reordering.

Audit completeness was preserved in all tested scenarios, with no missing or duplicated log entries observed during stress testing. Rollback safety was maintained through transaction ordering and cryptographic linkage of access records, ensuring that delayed blockchain commits do not invalidate previously enforced trust decisions. These results demonstrate that asynchronous logging is a viable deployment option for high-frequency IoT applications requiring ultra-low latency while retaining the integrity and accountability benefits of blockchain-based auditing.

Table 19 compares the latency and consistency characteristics of synchronous and asynchronous blockchain logging modes in AAZTC. While synchronous logging ensures strong consistency by committing access decisions before confirmation, it introduces higher decision latency. In contrast, asynchronous logging significantly reduces access decision latency by removing blockchain operations from the critical path, while maintaining full audit completeness and rollback safety through eventual consistency. These results demonstrate that asynchronous logging provides an effective trade-off between ultra-low latency requirements and blockchain-based accountability in high-frequency IoT applications.

Table 19 Latency and Consistency Comparison between Synchronous and Asynchronous Blockchain Logging

Mode	Mean Decision Latency (ms)	Blockchain Commit Delay (ms)	Audit Completeness	Consistency Model	Rollback Safety
Synchronous Logging	12.4	7.9	100%	Strong consistency	Guaranteed
Asynchronous Logging	4.6	18.3 (batched)	100%	Eventual consistency	Guaranteed

Scalability Extrapolation Beyond 10 Validators:

While experimental evaluation in Table 20 considers validator sizes up to 10 peers, the scalability behavior of the proposed AAZTC blockchain layer can be analytically extrapolated based on the communication complexity of the underlying PBFT-style consensus. PBFT incurs a message complexity of $O(V^2)$, leading to quadratic growth in consensus overhead as the number of validator peers increases. Using the measured latency at $V=10$ as a reference point, we extrapolate latency and throughput trends for larger validator sets ($V=30,50,100$).

The extrapolated results indicate that consensus latency increases non-linearly beyond 30 validators, while throughput degrades proportionally due to increased message exchanges. Despite this, access decision enforcement in AAZTC remains unaffected because trust evaluation and anomaly detection are executed locally and are decoupled from blockchain confirmation through asynchronous logging. Therefore, blockchain scalability primarily impacts audit confirmation delay rather than real-time trust enforcement.

Table 20 Projected Scalability of AAZTC under Increasing Validator Counts

Number of Validators (V)	Consensus Latency (ms)	Throughput (tx/s)	Consistency Model	Security Guarantees
10 (measured)	7.9	5,000	Strong consistency	Byzantine fault tolerant
30 (extrapolated)	21.4	3,200	Strong consistency	Byzantine fault tolerant
50 (extrapolated)	46.8	1,900	Strong consistency	Byzantine fault tolerant
100 (extrapolated)	121.6	850	Strong consistency	Byzantine fault tolerant

Discussion on Alternative BFT Protocols:

To mitigate the quadratic communication overhead of PBFT in large-scale IoT deployments, asynchronous BFT protocols such as HotStuff can be considered as future extensions of AAZTC. HotStuff reduces message complexity to $O(V)$ per consensus round while preserving Byzantine fault tolerance under partial synchrony. Under HotStuff-style consensus, blockchain commit latency would scale linearly with the number of validators, enabling deployments with tens or hundreds of validator nodes without compromising security guarantees. Importantly, the trust enforcement logic of AAZTC remains independent of the consensus mechanism, allowing seamless replacement of PBFT with more scalable BFT variants while preserving auditability, rollback safety, and access accountability.

From a system-level perspective, the quadratic communication overhead of PBFT does not impact real-time access decision-making in AAZTC. Trust evaluation, anomaly detection, and policy enforcement are executed locally at the Policy Enforcement Point, while blockchain consensus is invoked asynchronously for audit logging. As a result, increases in validator count primarily affect audit confirmation delay rather than enforcement latency. The extrapolated scalability results and comparison with lightweight BFT alternatives demonstrate that AAZTC maintains real-time responsiveness under PBFT for moderate validator sizes and can seamlessly adopt more scalable consensus protocols for larger deployments.

5.3 Deployment Assumptions and Realism

The AAZTC framework considers some of the practical constraints that are involved in the deployment of the IoT. Computing trust score and detecting anomalies is done with edge locations or at gateways to prevent constant transmission of raw data to central servers to minimize bandwidth usage and maintain privacy. Deep reinforcement learning model can be trained on a low-dimensional representation of states and the action space is limited allowing new edge-class inferences in real-time. Permissioned blockchain layer is expected to be run by infrastructure level devices like gateways, fog nodes (or organizational servers) instead of resource intensive IoT devices. The validator nodes are thus not supposed to be fully adversarial or highly resource constrained as this can be seen in real world enterprise and industrial IoT deployments. As much as the existing set ups of the experiment can test a small number

of validators, the architecture can scale horizontally by adding an extra validator nodes and asynchronous transaction processing, such that access decisions can be made in region, and audit records can be written to the ledger in parallel.

5.4 Security Analysis

The security variables of AAZTC are developed as a result of the synthesis of several defensive layers.

The blockchain aspect has integrity and non-repudiation assurances of access logs. The maximum number of nodes $[(V - 1)/3]$ is guaranteed by the PBFT consensus. Byzantine authenticators are unable to poison the registry. The unalterable audit trail allows the forensic investigation and compliance with regulations.

The post-quantum cryptography module based on LWE scientists react to the hardness of the Learning With Errors problem with quantum resistance and security lowering. Recent quantum algorithms provide no infantile-time solutions to LWE, which guarantees confidentiality over time, should other quantum adversaries, evident in the far off future.

The adaptive trust mechanism offers resilience to various forms of gradual trust exploitation attack where attackers gradually establish trust and then initiate attacks. The DRA learner gets used to ignoring any suspicious trust trajectory pattern and can prematurely lower the level of trustworthiness to the devices with precursor behaviors.

Rationale for Permissioned Blockchain Selection

AAZTC adopts a permissioned blockchain to balance zero-trust enforcement with real-time operational constraints in IoT environments. While permissionless blockchains provide trustless participation, they introduce nondeterministic confirmation latency, transaction fees, and limited policy control, which are incompatible with time-critical access decisions. In contrast, a permissioned blockchain enables deterministic auditability, controlled validator participation, and seamless integration with policy enforcement engines, while still assuming no implicit trust in device behavior or access requests. This design aligns with the zero-trust principle of continuous verification while preserving low-latency enforcement and scalable deployment.

Interpretability of DRL-Based Trust Decisions

In addition to security guarantees, the interpretability of the deep reinforcement learning-based trust decisions is also considered in AAZTC. The trust update mechanism operates on a structured and semantically meaningful state representation composed of the previous trust value, anomaly score derived from behavioral analysis, historical trust evolution, and contextual risk indicators. The action-value function explicitly evaluates the long-term impact of each trust adjustment action, allowing access decisions to be traced back to dominant contributing factors such as abnormal behavior intensity or sustained trust degradation. This state-level transparency enables feature-level reasoning about why access is granted or denied, thereby improving trustworthiness and practical acceptability of adaptive trust decisions in security-critical scenarios.

5.4.1. Post-Quantum Security Advantage over DRL-Trust

DRL-Trust focuses exclusively on adaptive trust optimization using reinforcement learning but relies entirely on classical cryptographic primitives for access control and logging. Such primitives are vulnerable to quantum attacks based on Shor's algorithm, rendering trust records and access logs insecure in a post-quantum threat model. In contrast, AAZTC integrates a lightweight lattice-based post-quantum cryptographic module grounded in the Learning With Errors (LWE) problem, providing quantum-resilient protection for trust records, access decisions, and policy enforcement logs.

The design of AAZTC is further informed by prior AI-blockchain security frameworks that emphasize privacy-preserving trust evaluation and auditable decision-making in IoT and transportation systems [42]–[44]. From a security standpoint, this distinction is critical. While DRL-Trust can adapt trust scores dynamically, it cannot guarantee the long-term integrity or confidentiality of trust evidence once quantum adversaries become practical. AAZTC ensures that audit trails remain tamper-proof and cryptographically secure even under future quantum attacks. Importantly, post-quantum cryptographic operations in AAZTC are deliberately scoped to access-control transactions and blockchain logging, avoiding continuous cryptographic processing during trust inference. Experimental results in Section 4.9 demonstrate that PQ operations introduce less than 2% additional latency, confirming that quantum resilience is achieved without compromising real-time zero-trust enforcement. This positions AAZTC as a future-proof zero-trust architecture, whereas DRL-Trust remains vulnerable to cryptographic obsolescence.

5.5 Comparison with Related Work

AAZTC addresses the core issue of being fixed: Unlike traditional implementations of zero-trust where access is decided at a single point (which is often called zero trust or static zero-trust) [4], adaptive trust boundaries are offered by AAZTC to evolve with device patterns. In contrast to attribute-based models [3], our behavioral analysis also is able to capture dynamic models that cannot be modeled using attribute based systems.

In comparison to federated learning methods [7], [10], AAZTC is not based on distributed model training, which makes it easier to deploy, and privacy through local trust computing. The blockchain layer offers similar auditability advantage wherein federated aggregation is not required.

Compared to pure DRA-based solutions [13], [14], AAZTC introduces post-quantum security and an entire blockchain integration, as the current solutions are still susceptible to quantum attacks and cannot give full audit facilities.

Compared to DRL-Trust, AAZTC advances adaptive trust management in three fundamental dimensions. First, AAZTC integrates anomaly-aware behavioral modeling into the reinforcement learning state, enabling robust detection of stealthy and slow-evolving attacks that DRL-Trust fails to capture. Second, AAZTC implements closed-loop zero-trust enforcement aligned with NIST SP 800-207, whereas DRL-Trust performs trust updates without explicit policy execution or enforcement isolation. Third, AAZTC incorporates post-quantum cryptographic protection for trust records and access logs, ensuring long-term security guarantees absent in DRL-Trust. These combined factors explain the superior performance results observed in Sections 4.4 and 4.7 and justify the architectural advantages beyond PQ security alone.

Table 21 summarizes the key architectural and security differences between DRL-Trust and AAZTC. While both frameworks employ reinforcement learning for adaptive trust computation, AAZTC extends this capability through anomaly-aware state modeling, closed-loop policy enforcement, and post-quantum cryptographic protection. These additions provide stronger security guarantees and explain the superior performance and robustness of AAZTC observed in the experimental results.

Table 21 Architectural and Security Comparison between DRL-Trust and AAZTC

Dimension	DRL-Trust	AAZTC
Trust adaptation mechanism	DRL-based	DRL-based
Anomaly-aware state modeling	No	Yes
Closed-loop policy enforcement	No	Yes

Blockchain audit logging	Yes	Yes
Logging mode	Synchronous	Asynchronous
Post-quantum cryptography	No	Yes (LWE-based)
Quantum attack resilience	No	Yes
NIST SP 800-207 compliance	Partial	Explicit

5.6 Limitations and Future Directions

The limitations of the present work highlight several important directions for future research. First, although the AAZTC framework is designed to operate in edge-assisted IoT environments, the combined computational overhead of deep reinforcement learning, anomaly detection, blockchain interaction, and post-quantum cryptography may be prohibitive for highly resource-constrained devices. This limitation can be mitigated through lightweight model compression, quantization, and hardware acceleration, which remain promising directions for deployment on ultra-low-power IoT nodes.

Second, the experimental evaluation relies on widely used benchmark datasets, which, while comprehensive, may not fully capture the operational complexity, noise, and heterogeneity of real-world IoT deployments. Factors such as device churn, intermittent connectivity, and evolving traffic patterns are difficult to model in offline datasets. Future work will focus on validating AAZTC in real-world or large-scale testbed environments to better assess robustness under practical conditions.

Third, the blockchain layer introduces inherent latency due to consensus requirements. Although asynchronous logging significantly reduces decision latency, applications with ultra-strict real-time constraints may still find blockchain-based audit confirmation unsuitable. Exploring faster and more scalable consensus mechanisms, such as asynchronous Byzantine fault-tolerant protocols, represents a key direction for improving system responsiveness while maintaining auditability.

Fourth, the current threat model assumes a semi-trusted deployment environment, where validators are honest-but-curious and fewer than $\lfloor (V - 1)/3 \rfloor$ nodes may behave maliciously, consistent with PBFT assumptions. Stronger adversarial models involving adaptive or colluding validators beyond this threshold are not addressed and warrant further investigation. In addition, adversarial threats targeting both the learning and blockchain components are considered within the scope of AAZTC. Model poisoning and reward manipulation attacks against the deep reinforcement learning agent are partially mitigated through bounded trust score updates, anomaly-aware reward shaping, and continuous trust recalibration, which prevent abrupt trust escalation from corrupted observations. At the blockchain layer, large-scale attacks such as 51% attacks are mitigated by the permissioned deployment model and the Byzantine fault tolerance guarantees of PBFT, which assume fewer than $\lfloor (V - 1)/3 \rfloor$ malicious validators. While fully adaptive adversarial learning attacks and colluding validator scenarios beyond this threshold are not addressed in the current implementation, the modular architecture of AAZTC provides a foundation for incorporating adversarially robust learning methods and enhanced consensus hardening mechanisms in future extensions.

Finally, the autoencoder-based anomaly detection component may be vulnerable to evasion attacks in which adversaries gradually adapt their behavior to resemble benign traffic, thereby reducing reconstruction error over time. While reinforcement learning-based trust evolution helps mitigate abrupt trust escalation, future work should consider adversarially robust anomaly detection techniques and ensemble-based defenses to further strengthen resilience.

In addition, the current threat model does not explicitly address denial-of-service attacks, physical device compromise, or side-channel attacks, which are common in real-world IoT deployments. These attack classes primarily target system availability or hardware-level integrity rather than trust computation and policy enforcement logic. While their exclusion is intentional to maintain analytical focus, it limits the direct applicability of AAZTC in highly adversarial physical environments. In practical deployments, AAZTC can be complemented with network-level DoS mitigation mechanisms, physical tamper-resistance and secure boot technologies, and side-channel hardened cryptographic implementations. Such protections are orthogonal to the proposed trust and audit framework and represent important directions for extending AAZTC toward comprehensive end-to-end IoT security.

6. Conclusion

This paper introduced Adaptive-AI-ZeroTrust-Chain (AAZTC), a holistic view of security that combines the AI-driven models of dynamic trust boundaries with the verifiable access registry with blockchains to enforce specific quanta of zero-trust over IoT neighborhoods in a manner that was auditable. The framework uses mathematical algorithms of deep reinforcement learning to conduct continuous behavioral analysis and trust score calculation, uses smart contracts on a permissioned blockchain to conduct transparent policy execution, and uses lightweight lattice-based post-quantum cryptography to provide future resistance security. Commensive experiments with the NSL-KDD and CICIDS2017 datasets have shown that AAZTC attains 98.73% detection accuracy, 97.89% precision and 98.21% F1-score which are 3.2%-5.8% above state-of-the-art baselines. Although NSL-KDD is claimed to be backward compatible with the previous literature, the efficacy and actual utility of AAZTC are shown mainly by a single and comprehensive analysis on existing TON IoT and CICIDS2017 data sets. The system has low latency properties and the average time of trust decisions is 12.4 milliseconds and thus it fits the real-time IoT deployment. The synergistic performance of each architectural element was proved by ablation studies. Future is the exploration of lightweight model variants of resource-constrained devices, improved versions of consensus mechanisms, and long-term security analysis in the adversarial deployment conditions.

Appendix A: Nomenclature

Symbol	Description
$\mathcal{D}$	Set of IoT devices
d_i	Device i in the network
N	Total number of devices
$x_i(t)$	Behavioral feature vector of device i at time t
M	Number of extracted features
$H_i^{(p)}(t)$	Protocol distribution entropy
$\Delta f_i(t)$	Access frequency deviation
ϕ_{enc}	Autoencoder encoder function
ϕ_{dec}	Autoencoder decoder function
L	Latent dimension of autoencoder
$\mathcal{A}_i(t)$	Anomaly score for device i
$\tilde{\mathcal{A}}_i(t)$	Normalized anomaly score
$\mathcal{S}$	MDP state space
$\mathcal{A}$	MDP action space
$\mathcal{P}$	State transition probability function
$\mathcal{R}$	Reward function
$z\tau_i(t)$	Trust score of device i at time t
$Q(s, a; \theta)$	Q-function parameterized by θ

θ	Q-network parameters
θ^-	Target network parameters
$\mathcal{B}$	Experience replay buffer
$\mathcal{V}$	Set of blockchain validator nodes
V	Number of validators
T_x	Blockchain transaction
B_k	Block k in blockchain
H_k	Hash of block k
$T_{\text{consensus}}$	Consensus latency
δ	Network propagation delay
λ	Local processing time
$\tau_{\min}^{(j)}$	Minimum trust threshold for resource j
A	LWE public matrix
s	LWE secret key
e	LWE error vector
q	LWE modulus
n, m	LWE lattice dimensions
Acronym	Description
AAZTC	Adaptive-AI-ZeroTrust-Chain
AI	Artificial Intelligence
AUC	Area Under Curve
DQN	Deep Q-Network
DRL	Deep Reinforcement Learning
IoT	Internet of Things
LWE	Learning With Errors
MDP	Markov Decision Process
PBFT	Practical Byzantine Fault Tolerance
PQ	Post-Quantum
ROC	Receiver Operating Characteristic
SDN	Software-Defined Networking
SSI	Self-Sovereign Identity
ZT	Zero-Trust

Appendix B: PyTorch Implementation of the DQN Q-Network

```

import torch

import torch.nn as nn
import torch.nn.functional as F

class QNetwork(nn.Module):
    """
    Deep Q-Network for trust score adjustment in AAZTC.
    Architecture: Input -> 256 -> 128 -> 64 -> Output(5 actions)
    """
    def __init__(self, state_dim: int, action_dim: int = 5):
        super().__init__()
        self.fc1 = nn.Linear(state_dim, 256)
        self.fc2 = nn.Linear(256, 128)
        self.fc3 = nn.Linear(128, 64)
        self.out = nn.Linear(64, action_dim)

        # Optional but common: stable initialization
        nn.init.kaiming_uniform_(self.fc1.weight, nonlinearity="relu")

```

```

nn.init.kaiming_uniform_(self.fc2.weight, nonlinearity="relu")
nn.init.kaiming_uniform_(self.fc3.weight, nonlinearity="relu")
nn.init.xavier_uniform_(self.out.weight)

```

```

def forward(self, x: torch.Tensor) -> torch.Tensor:
    x = F.relu(self.fc1(x))
    x = F.relu(self.fc2(x))
    x = F.relu(self.fc3(x))
    return self.out(x)

```

Appendix B : Dynamic Trust Score Update via Reinforcement Learning

The following code snippet demonstrates the practical implementation of dynamic trust score adjustment using reinforcement learning as described in Section III. The function `update_trust_score_with_drl()` shows how behavioral observations and anomaly signals are transformed into trust score updates through DRL-based action selection, corresponding to Equations (6), (7), and (14) in the methodology section.

```

def update_trust_score_with_drl(device_id, behavioral_features, anomaly_score,
                                q_network, trust_scores, epsilon=0.1):

```

```

    """

```

Demonstrate dynamic trust score adjustment using DRL-based decision-making.

This function implements the core trust evolution mechanism described in Section III, illustrating how reinforcement learning enables adaptive zero-trust enforcement based on continuous behavioral analysis.

Args:

`device_id`: Unique identifier for the IoT device
`behavioral_features`: Extracted behavioral feature vector $x_i(t)$
`anomaly_score`: Normalized anomaly score $A_i(t)$ from autoencoder
`q_network`: Trained Deep Q-Network with parameters θ
`trust_scores`: Dictionary mapping `device_id` to current trust score $\tau_i(t)$
`epsilon`: Exploration rate for ϵ -greedy action selection

Returns:

tuple: (updated_trust_score, selected_action, trust_adjustment)

```

    """

```

```

import torch

```

```

import numpy as np

```

```

# Step 1: Retrieve current trust score  $\tau_i(t)$ 

```

```

current_trust = trust_scores.get(device_id, 0.5) # Default: neutral trust

```

```

# Step 2: Construct state representation  $s_t$  (Equation 6)
# State = [previous_trust, anomaly_score, historical_features, context]
historical_trend = np.mean(behavioral_features[-10:]) if len(behavioral_features) >= 10 else
np.mean(behavioral_features)
current_context = behavioral_features[-1] if len(behavioral_features) > 0 else 0.0

state = torch.FloatTensor([
    current_trust,      #  $\tau_i(t)$ 
    anomaly_score,      #  $A_i(t)$ 
    historical_trend,    #  $H_i(t)$ 
    current_context     #  $C_i(t)$ 
])

# Step 3:  $\epsilon$ -greedy action selection (Equation 7)
if np.random.random() < epsilon:
    # Exploration: random action from  $A = \{a_0, a_1, a_2, a_3, a_4\}$ 
    action = np.random.randint(0, 5)
else:
    # Exploitation: select action with highest Q-value
    #  $Q(s_t, a; \theta)$  for all  $a \in A$ 
    with torch.no_grad():
        q_values = q_network(state.unsqueeze(0))
        action = torch.argmax(q_values).item()

# Step 4: Map discrete actions to trust score adjustments
# Actions:  $a_0=-0.2, a_1=-0.1, a_2=0.0, a_3=+0.1, a_4=+0.2$ 
delta_trust = [-0.2, -0.1, 0.0, 0.1, 0.2][action]

# Step 5: Update trust score with bounds enforcement (Equation 14)
#  $\tau_i(t+1) = \text{clip}(\tau_i(t) + \delta(a_t), 0, 1)$ 
updated_trust = np.clip(current_trust + delta_trust, 0.0, 1.0)

# Step 6: Store updated trust score
trust_scores[device_id] = updated_trust

return updated_trust, action, delta_trust

```

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